

Spacecraft Attitude Determination and Control Technical Report

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Change Log

Homework	Date	Description of Changes
1	02-27-2026	Described mission description, requirements and spacecraft model. Calculated COM, body inertia, principle axes, surface normal vectors and surface areas of the spacecraft. Simulated spacecraft orbital dynamics and Euler's equation. Provided a stability analysis about each principle axis with plots on the momentum sphere.
2	03-13-2026	Designed a stable spin configuration that keeps the solar panels pointing toward the sun using super spin and dynamic balance. Added gyrostat dynamics to the simulation and described attitude sensors. Simulated noisy measurements from each attitude sensor and used these simulated measurements for static attitude determination.
3	04-07-2026	Fixed dynamic balance bug (was not renormalizing the quaternion in the attitude dynamics sim). Organized the code base. Implement sensor models and a recursive attitude estimator.
4	05-01-2026	Add actuator specifications, environmental torques, an LQR controller for attitude regulation, and generate state and input trajectories for an eigen-axis slew maneuver.
5	05-13-2026	Add CBF CMG control case study

Github Repository: <https://github.com/vegaf1/acds-mit-course>

Homework 1

Mission Description and Requirements

This report analyzes the Starling mission. It consists of four 6-unit CubeSats flying in a sun synchronous circular orbit at an altitude of 550 km. The high-level mission objectives consist of swarm maneuver planning and execution, relative navigation, and autonomous coordination between the spacecraft [1]. Onboard the satellites, there are two commercial star-trackers, dual-band GPS receivers, reaction wheels, a Globalstar beacon radio, and an S-band radio [2]. The actuation includes a cold gas propulsion system called Hamlet providing up to 30 m/s of delta-V for maneuvers [3]. The satellite is a Blue Canyon XB 6U cubesat which has a the pointing accuracy of ± 0.002 degrees, slew rate of 360 degrees per minute, and pointing stability of 1 arcsecond over 1 second [4].

Spacecraft Model

Part 1: We assume the dimensions of the 6U cubesat are 20 cm x 10 cm x 34 cm. Another assumption is the size of the panels are 30 cm x 100 cm (thickness not taken into account). The body axes are defined in Figure 1. The origin is defined to be the center of the spacecraft bus.

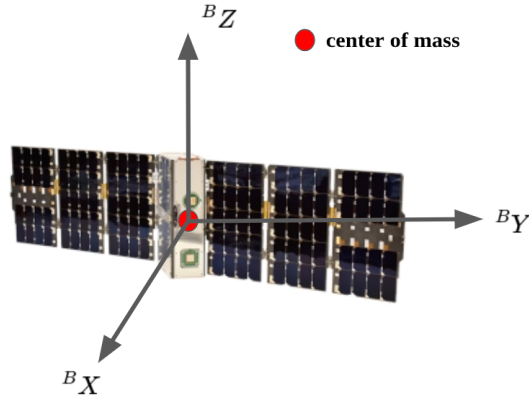
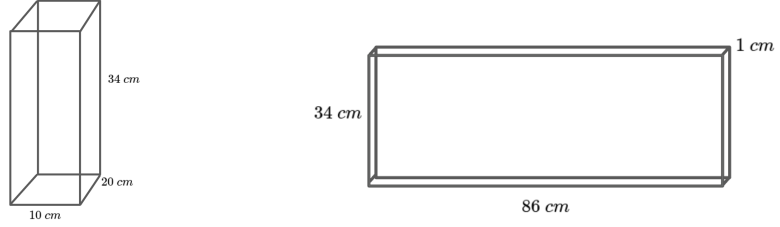


Figure 1: Body axes definition for Starling mission spacecraft

Part 2: We model the spacecraft as 3 rectangular prisms. One larger rectangular prism for the spacecraft body and another 2 thin rectangular prisms representing the solar panels. This decomposition along with the dimensions of each rectangular prism are depicted in Figure 2. To find the center of mass, we apply Equation 1, where m_k is the mass of each of the rectangular prisms and x_k is the distance from the body frame origin to the center of mass of each



(a) Starling bus rectangle decomposition and dimensions (b) Starling solar panels rectangular decomposition and dimensions

Figure 2: Starling 6U Cubesat Rectangular Decomposition and Dimensions

rectangular prism. We assume the mass of the satellite bus m_1 is 12 kg, the mass of each solar panel (m_2, m_3) is 1 kg. Another assumption is that the solar panels are at 45 degrees from the body X-axis and Y-axis.

$$X_{com} = \frac{\sum_{k=1}^n m_k x_k}{\sum_{k=1}^n m_k} \quad (1)$$

COM Calculation

$m_1 = 12 \text{ kg}$, $r_1 = (0, 0, 0) \text{ cm}$

$m_2 = 1 \text{ kg}$, $r_2 = (35.36, 35.36, 0) \text{ cm}$

$m_3 = 1 \text{ kg}$, $r_3 = (-35.36, -35.36, 0) \text{ cm}$

$r_{\text{com}} = [0, 0, 0] \text{ cm}$

This makes sense spacecraft because the spacecraft is symmetric.

To calculate the body-frame moment of inertia, we need to combine the individual inertias of each rectangular prism. We do this by adding the inertias of each individual term and adding the point-mass terms about the combined centered of mass (Equation 2).

$$J_{\text{combined}} = J_1 + J_2 + J_3 + m_1(r_1^T r_1 I - r_1 r_1^T) + m_2(r_2^T r_2 I - r_2 r_2^T) + m_3(r_3^T r_3 I - r_3 r_3^T) \quad (2)$$

In this equation, the term r_k is the distance from each rectangular prism center of mass to the center of mass of the spacecraft. Since the COM of the spacecraft is the origin of the body frame, the r terms remain the same from

the previous part. ${}^B J = \begin{bmatrix} 5915.99 & -2500.66 & 0.0 \\ -2500.66 & 3949.49 & 0.0 \\ 0.0 & 0.0 & 7168.15 \end{bmatrix} \text{ kg} \cdot \text{cm}^2$

Part 3: The principle axes of a spacecraft is the basis where the body frame moment of inertia is diagonal. We calculate by taking an eigen decomposition of the body frame moment of inertia. The set of eigenvectors becomes the principle axes P .

$$P = \begin{bmatrix} 0.56306 & 0.0 & 0.826416 \\ 0.826416 & 0.0 & -0.56306 \\ 0.0 & 1.0 & 0.0 \end{bmatrix}$$

The rotation between the principle frame and body frame is defined by matrix P . To transform the body frame inertia into the principle frame inertia, we apply the following: ${}^P J = P^T B J P$.

Part 4: We can discretize the satellite bus into 6 faces, and the solar panels into 5 faces each. This is a total of 16 faces to represent the spacecraft. The next section* will specify the centroid, normal vector, and area of each piece.

Spacecraft bus:

Face 1:

Centroid: $[0, 5, 0] \text{ cm}$

Normal Vector: $[0, 1, 0]$

Area: 680 cm^2

Face 2:

Centroid: $[0, -5, 0] \text{ cm}$

Normal Vector: $[0, -1, 0]$

Area: 680 cm^2

Face 3:

Centroid: $[10, 0, 0] \text{ cm}$

Normal Vector: $[1, 0, 0]$

Area: 340 cm^2

Face 4:

Centroid: $[-10, 0, 0]$ cm

Normal Vector: $[-1, 0, 0]$

Area: 340 cm^2

Face 5:

Centroid: $[0, 0, 17]$ cm

Normal Vector: $[0, 0, 1]$

Area: 200 cm^2

Face 6:

Centroid: $[0, 0, -17]$ cm

Normal Vector: $[0, 0, -1]$

Area: 200 cm^2

Solar Panel 1:

Face 1:

Centroid: $[35.36, 35.36, 0]$ cm

Normal Vector: $[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$

Area: 2924 cm^2

Face 2:

Centroid: $[35.36, 35.36, 0]$ cm

Normal Vector: $[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$

Area: 2924 cm^2

Face 3:

Centroid: $[35.36, 35.36, 17.0]$

Normal Vector: $[0, 0, 1]$

Area: 86 cm^2

Face 4:

Centroid: $[35.36, 35.36, -17.0]$

Normal Vector: $[0, 0, -1]$

Area: 86 cm^2

Face 5:

Centroid: $[70.71, 70.71, 0.0]$

Normal Vector: $[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$

Area: 86 cm^2

Solar Panel 2:

Face 1:

Centroid: $[-35.36, -35.36, 0]$ cm

Normal Vector: $[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]$

Area: 2924 cm^2

Face 2:

Centroid: $[-35.36, -35.36, 0]$ cm

Normal Vector: $[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$

Area: 2924 cm^2

Face 3:

Centroid: $[-35.36, -35.36, 17.0]$

Normal Vector: $[0, 0, 1]$

Area: 86 cm^2

Face 4:

Centroid: [-35.36, -35.36, -17.0]

Normal Vector: [0, 0, -1]

Area: 86 cm²**Face 5:**

Centroid: [-70.71, -70.71, 0.0]

Normal Vector: $[-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]$ Area: 86 cm²

Dynamics

Equation 3 calculates the gravitational acceleration, and Equation 4 calculates the J2 acceleration. The variable r is the position and v is the velocity. For now, we assume that the total acceleration is the sum of both of these accelerations.

$$a_g = \frac{-\mu}{||r||^3} r \quad (3)$$

$$a_{J_2} = -\frac{3}{2} \frac{J_2 \mu R_E^2}{r^5} \begin{bmatrix} \left(1 - 5 \left(\frac{x_3}{r}\right)^2\right) x_1 \\ \left(1 - 5 \left(\frac{x_3}{r}\right)^2\right) x_2 \\ \left(3 - 5 \left(\frac{x_3}{r}\right)^2\right) x_3 \end{bmatrix} \quad (4)$$

$$a_{\text{total}} = a_g + a_{J_2} \quad (5)$$

We implemented an RK4 integrator and simulated the dynamics for 10 revolutions. The resulting trajectory is in Figure 3.

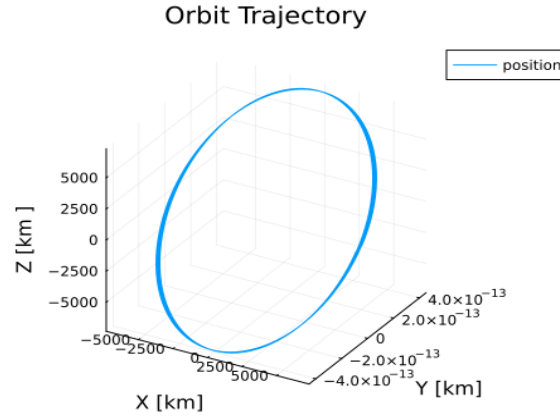


Figure 3: Orbit Trajectory

Euler's Equation

Part 1: The ODE's for Euler's equation along the principle axes are in Equation 6.

$$\begin{aligned} {}^P J_{11} {}^P \dot{\omega}_1 + ({}^P J_{33} - {}^P J_{22}) {}^P \omega_2 {}^P \omega_3 &= {}^P \tau_1 \\ {}^P J_{22} {}^P \dot{\omega}_2 + ({}^P J_{11} - {}^P J_{33}) {}^P \omega_1 {}^P \omega_3 &= {}^P \tau_2 \\ {}^P J_{33} {}^P \dot{\omega}_3 + ({}^P J_{22} - {}^P J_{11}) {}^P \omega_1 {}^P \omega_2 &= {}^P \tau_3 \end{aligned} \quad (6)$$

Part 2: The initial condition simulated had a magnitude of 10 RPM. The perturbation had a magnitude of $0.1 \frac{\text{rad}}{\text{s}}$ and the direction was sampled from a normal distribution. Figure 4 shows a spin around the minor axis that is marginally stable. There is nutation around the y and z axes of the principle frame. Figure 6 shows a spin around the major axis that is also marginally stable. There is nutation around the x and z principle axes. Finally, Figure 5 shows an unstable spin around the intermediate axis. We can observe the switching behavior that occurs in the x and y-directions of the angular velocity from positive to negative.

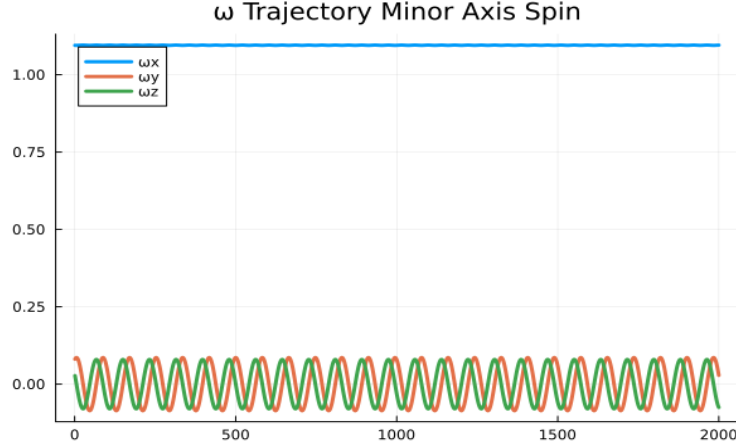


Figure 4: Minor Axis Spin Angular Velocity Trajectory

Part 3: Figures 7 and 8 show a plot of the momentum sphere with the equilibrium points. The green markers indicate the stable equilibrium points (minor/major axis) and the red markers are the unstable equilibria (intermediate axis).

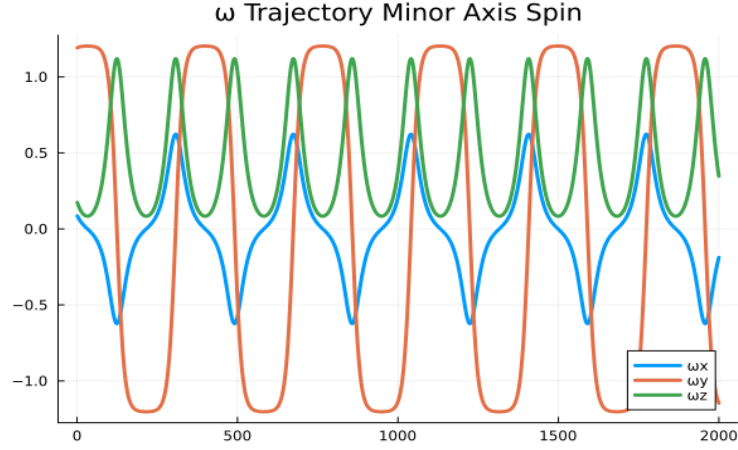


Figure 5: Intermediate Axis Spin Angular Velocity Trajectory

Next, we show several momentum trajectories on the momentum sphere from spinning about each of the principle axes. The momentum trajectories are depicted by the blue lines while the equilibrium points are represented by markers. The red marker indicates unstable equilibria (along the intermediate axis), while green are the stable equilibria. For the intermediate axis, the initial condition has a magnitude of 20 RPM with a perturbation in a random direction with a magnitude of $0.1 \frac{rad}{s}$. The momentum trajectory about the intermediate axis is shown in Figure 9. It shows the same switching behavior we observed from the angular velocity plot in Figure 5

Next, we simulated a spin about the minor axis. The initial angular velocity is set to a magnitude of 3 RPM with a disturbance in a random direction with a magnitude of $0.1 \frac{rad}{s}$. The momentum trajectory about the minor axis is shown in Figure 10. This is a stable equilibria, therefore the momentum trajectory is a periodic orbit about the equilibrium point.

Lastly, we simulated a spin about the major axis with an initial angular velocity magnitude set to 3 RPM with an additive random disturbance with a magnitude of $0.1 \frac{rad}{s}$. This is also a stable equilibria; therefore, the result is a periodic orbit about the equilibrium point.

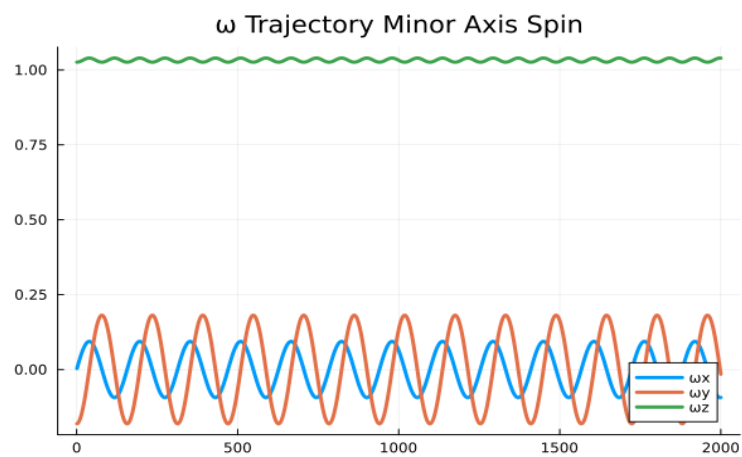


Figure 6: Major Axis Spin Angular Velocity Trajectory

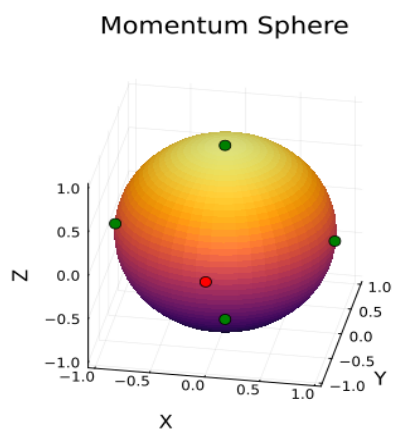


Figure 7: Momentum Sphere View 1

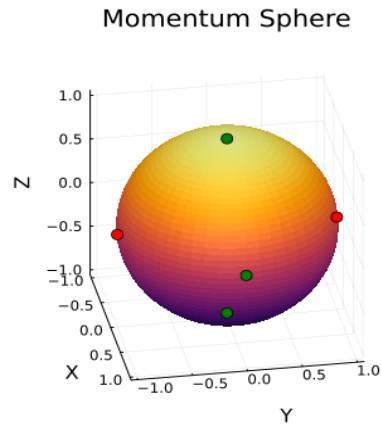


Figure 8: Momentum Sphere View 2

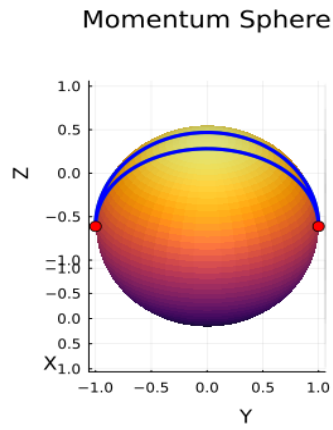


Figure 9: Momentum Trajectory about the Intermediate Axis

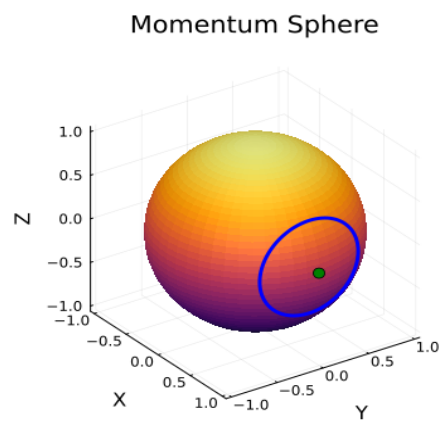


Figure 10: Momentum Trajectory about the Minor Axis

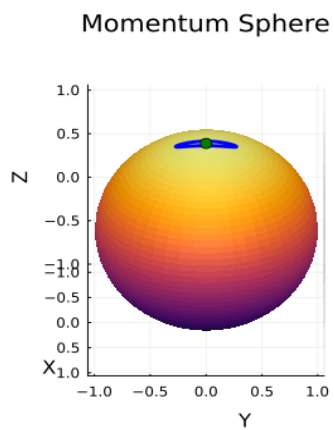


Figure 11: Momentum Trajectory about the Major Axis

Homework 2

Safe Mode

1. To ensure the solar panel normal vector is not parallel to a principle axis, we randomly perturb the body inertia matrix. This is done by taking an eigen-decomposition, perturbing the diagonal vector containing the eigenvalues, and reconstructing the body inertia matrix using the perturbed eigenvalue matrix.
2. To find the desired angular velocity vector in the body frame corresponding to a spin of 10 RPM about the solar panel vector we multiply the magnitude by the direction of the solar panel normal in the body frame. ${}^B\omega = ||\omega||^B n$

$${}^B\omega = \begin{bmatrix} -7.07 \\ 7.07 \\ 0 \end{bmatrix} \text{ rad/s}$$

3. Dynamic balance allows us to spin a spacecraft along a non-principle axis and superspin allows the spacecraft obtain a marginally stable spin by changing it's major axis by adding a rotor momentum to set an effective inertia greater than the major axis inertia by some margin. Using dynamic balance and superspin, we are able to stabilize to spin the satellite about the angular velocity vector in the direction of the solar panel normal. For dynamic balance, $\dot{\omega} = 0$, therefore from the gyrostat equation, we define the no wobble condition as the following

$$\omega \times (J\omega + \rho) = 0 \quad (7)$$

To obtain a unique solution, we also define along with the super spin condition $\rho^T \omega = \rho_s \omega_s$ to get a unique solution. The values ($\omega_s = ||\omega||$ and ρ_s are obtained from the superspin condition $\rho_s \geq \omega_s(J_{33} - J_s)$). We choose J_{33} here because it is the minimum-energy state. The term J_s is the inertia in the desired spin axis frame $J_s = (\frac{\omega}{\omega_s})^T J (\frac{\omega}{\omega_s})$.

4. The gyrostat equation is

$$J^B \omega + {}^B \dot{\rho} + {}^B \omega \times (J^B \omega + \rho) = {}^B \tau \quad (8)$$

where J is the inertia in the body frame, ${}^B\omega$ is the angular velocity in the body frame, ρ is the rotor momentum, and $\dot{\rho}$ is the rotor momentum rate.

5. The perturbation on the initial state has a magnitude of $0.1 \frac{\text{rad}}{\text{s}}$ and sampled from a normal distribution. The result of the simulation is stable about the solar panel normal vector as shown in Figure 12.

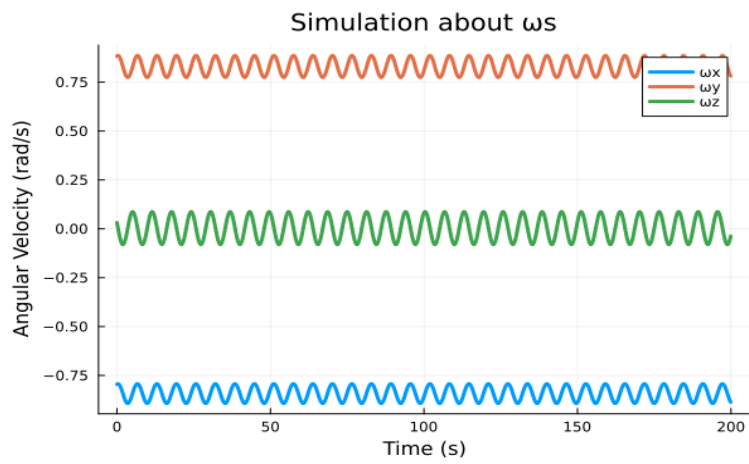


Figure 12: Angular velocity simulation about the solar panel normal vector

Spacecraft Dynamics

Figure 13 shows the attitude quaternion components for a stable spin about the solar panel normal vector. The simulation rate is set to 10 Hz and the horizon is set to approximately 3 minutes (2000 timesteps). Figure 14 shows the pointing error in degrees from the solar panel normal vector and the sun vector that is assumed to be at $[1,0,0]$ in ECI.

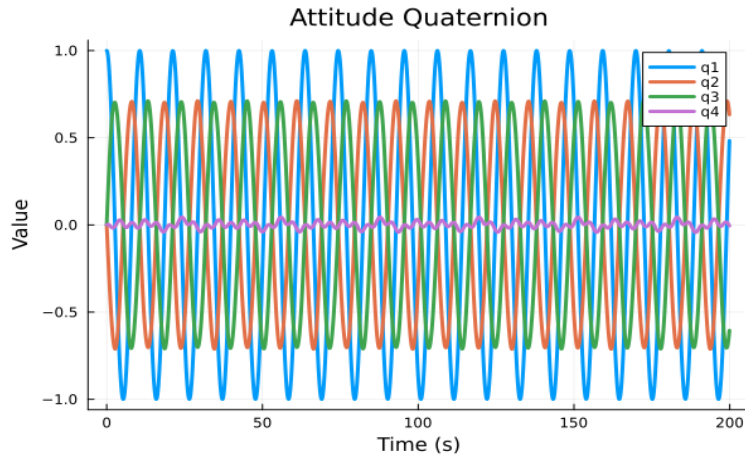


Figure 13: Components of the Attitude Quaternion

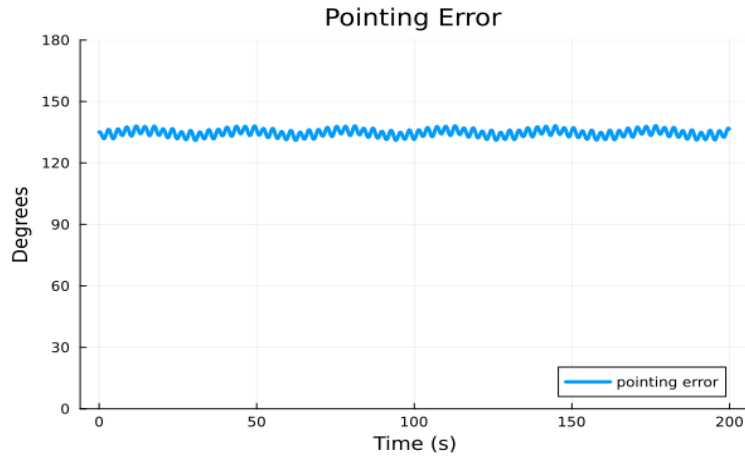


Figure 14: Pointing Error with the sun vector

Attitude Sensors

The attitude sensors onboard the spacecraft are two commercial star trackers. The reported pointing accuracy is ± 0.002 degrees as the $1\text{-}\sigma$ standard deviation [1]. From this specification we must convert the angular metric into an uncertainty on a bearing vector. We define a bearing measurement at time i as b_i . The covariance is dependent on the direction of the bearing measurement, as the uncertainty will be in the orthogonal subspace to b_i . To do this, we can use the projection $(I - b_i b_i^T)$ and multiply it by our variance converted into radians (this is valid for small angles, which is the case for this sensor). The equation for the covariance matrices is

$$P_s = \sigma^2(I - b_i b_i^T) \quad (9)$$

We can then sample a noisy measurement b_n by

$$b_n = b_i + \sqrt{P_s}d \quad (10)$$

where d is a 3 dimensional vector sampled from a normal distribution. To verify that the error statistics are consistent, we can take the arc cosine of the dot product between the normal body measurement and the true body measurement. This operation gives us the angle difference between both vectors and most should fall within the 1-sigma range. As seen in Figure 15 the average angle difference is close to the 0.002 deg metric from the star tracker specification sheet.

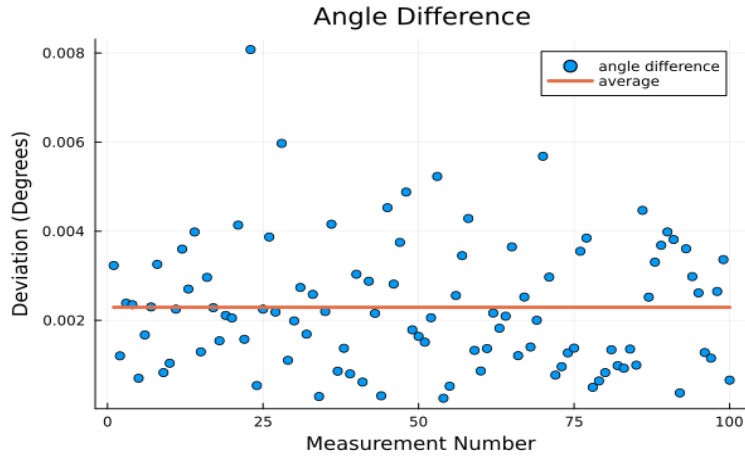


Figure 15: Angle difference between the noisy measurements and true measurements in the body frame

Static Attitude Estimation

For 100 measurements, we ran a MonteCarlo simulation of both the SDP and SVD method of solving Wabha's problem. The rotation error in degrees are very similar and shown in Figures 16 and 17. Using BenchmarkTools.jl in Julia, we are also able to obtain an average time for each approach to run. The average time for the SDP method was 0.00184 seconds while the SVD solution ran in 7.007×10^{-6} seconds.

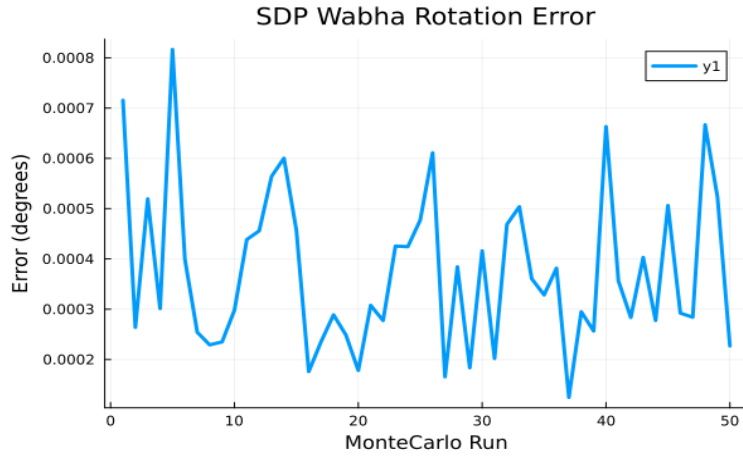


Figure 16: Rotation Error in degrees from solving Wabha's problem using an SDP

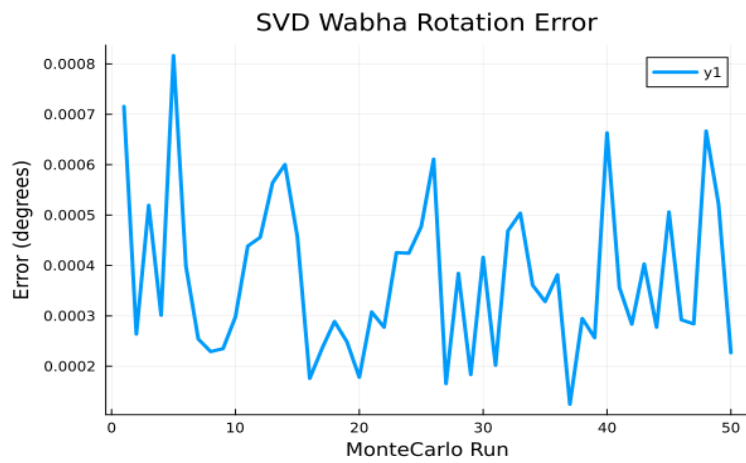


Figure 17: Rotation Error in degrees from solving Wabha's problem using an SVD

Homework 3

Attitude Sensors

The Blue Canyon XB6 satellites are equipped with a custom attitude determination and control system called XB1 [1]. The attitude sensors on the module are a MEMS magnetometer, MEMS IMU, sun sensors, and nano star trackers [5]. Since one of the requirements of Starling mission was to demonstrate angles-only navigation, it is equipped with two star trackers aligned in anti-parallel directions [6]. We will assume the type of sun sensors, magnetometers, and IMU used since the specific versions are not specified in the literature. We obtain accurate characterizations of each sensor through their data sheet. For small angles, we can define a misalignment matrix M [7] where

$$M = \begin{bmatrix} 1 & -\Delta\theta_z & \Delta\theta_y \\ \Delta\theta_z & 1 & -\Delta\theta_x \\ -\Delta\theta_y & \Delta\theta_x & 1 \end{bmatrix} \quad (11)$$

and $\Delta\theta_x, \Delta\theta_y, \Delta\theta_z$ are the misalignment errors in radians. Both the sun sensor and magnetometer measurements will have the following form

$$y = My_{ideal} + b + w, \quad w \sim N(0, W) \quad (12)$$

where y is the noisy measurement, b is the constant bias vector, and w is the additive noise sampled from a normal distribution with covariance W .

1. We assume that the module has an Aquila sun sensor from NewSpace Systems. The data sheet reports an average accuracy of 0.5 degrees. Therefore, we assume an alignment error of 0.1 degrees in each axis, a constant bias of 0.05 degrees, and a 1-sigma standard deviation for the additive noise at 0.5 degrees [8]. Typically, sun sensors detect the cosine of the angle between the sun and the sensor and the angle is determined through Newton's method [9]. For this simulation, we assume we have the sun vector with respect to the spacecraft body frame.

$$M = \begin{bmatrix} 1 & -1.74 \times 10^{-3} & 1.74 \times 10^{-3} \\ 1.74 \times 10^{-3} & 1 & -1.74 \times 10^{-3} \\ -1.74 \times 10^{-3} & 1.74 \times 10^{-3} & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 8.72 \times 10^{-4} \\ 8.72 \times 10^{-4} \\ 8.72 \times 10^{-4} \end{bmatrix},$$

$$W = \begin{bmatrix} 7.61 \times 10^{-5} & 0 & 0 \\ 0 & 7.61 \times 10^{-5} & 0 \\ 0 & 0 & 7.61 \times 10^{-5} \end{bmatrix} \quad (13)$$

The typical alignment for a MEMS magnetometer is ± 1 degrees and the standard deviation in each axis for the additive noise w is set to 0.4 mG (milliGauss) which is equivalent to 40 nanoTeslas. We set the constant bias to 3 mG which

is 100 nanoTeslas [10].

$$M = \begin{bmatrix} 1 & -1.74 \times 10^{-2} & 1.74 \times 10^{-2} \\ 1.74 \times 10^{-2} & 1 & -1.74 \times 10^{-2} \\ -1.74 \times 10^{-2} & 1.74 \times 10^{-2} & 1 \end{bmatrix} rad, \quad b = \begin{bmatrix} 100 \\ 100 \\ 100 \end{bmatrix} nT,$$

$$W = \begin{bmatrix} 1600 & 0 & 0 \\ 0 & 1600 & 0 \\ 0 & 0 & 1600 \end{bmatrix} (nT)^2 \quad (14)$$

2. For the star trackers, the 1-sigma standard deviation for bore-sight accuracy is 4 arcseconds and the roll-axis is 30 arcseconds [5]. This error is modeled as random rotation applied to the quaternion measurement and will be sampled from a Gaussian defined over axis-angle vectors. We assume the Z-axis in the body frame is the boresight direction, therefore the covariance matrix W in which we sample the axis angle vector is defined as

$$W = \begin{bmatrix} 3.76 \times 10^{-10} & 0 & 0 \\ 0 & 3.76 \times 10^{-10} & 0 \\ 0 & 0 & 2.12 \times 10^{-8} \end{bmatrix} rad^2 \quad (15)$$

$$\phi \sim \mathcal{N}(0, W) \quad (16)$$

We then apply the quaternion exponential map to the axis angle vector and left multiply the result by the ground truth quaternion q .

$$q_{ST} = L(q)expq(\phi) \quad (17)$$

Since we have 2 star trackers on board, we obtain 2 quaternion estimates.

3. We assume a MEMS IMU from Analog Devices [11]. From the data sheet, the misalignment error is ± 0.25 degrees, the bias modeled as a random walk has a 1-sigma standard deviation of $[0.13, 0.13, 0.19]$ degrees per $\sqrt{hr}$, and the 1-sigma of the noise is set at $[1.5, 2.3, 1.7]$ degrees per hour for the xyz axes. The covariance matrix for the bias vector is W_b , the misalignment matrix is M , and the gyroscope measurement covariance is defined by W .

$$M = \begin{bmatrix} 1 & -4.36 \times 10^{-2} & 4.36 \times 10^{-2} \\ 4.36 \times 10^{-2} & 1 & -4.36 \times 10^{-2} \\ -4.36 \times 10^{-2} & 4.36 \times 10^{-2} & 1 \end{bmatrix} rad, W_b = \begin{bmatrix} \frac{5.15 \times 10^{-6}}{\sqrt{dt}} & 0 & 0 \\ 0 & \frac{5.15 \times 10^{-6}}{\sqrt{dt}} & 0 \\ 0 & 0 & \frac{1.10 \times 10^{-5}}{\sqrt{dt}} \end{bmatrix} (rad/hr)^2$$

$$W = \begin{bmatrix} 6.85 \times 10^{-4} & 0 & 0 \\ 0 & 1.68 \times 10^{-3} & 0 \\ 0 & 0 & 8.80 \times 10^{-4} \end{bmatrix} (rad/hr)^2 \quad (18)$$

The gyroscope measurement model defined as

$$y_{gyro} = M\omega_{true} + b(t) + w \quad (19)$$

where w is sampled from covariance W and $b(t)$ is the time varying bias that is modeled as a random walk with covariance W_b .

Recursive Attitude Estimation

We simulate the satellite in an uncontrolled tumble by sampling an initial angular velocity (ω_0) from a Gaussian distribution with 1-sigma standard deviation of $3.0 \frac{\text{deg}}{\text{s}}$.

1. The maximum slew rate of the spacecraft is 360 degrees per minute which is 6 degrees per second. Therefore, the rate of the sample rate of the filter must be small enough to run multiple steps before hitting the maximum rotation as well as small enough to assume that the angular velocity is constant over a sample period. We select running the filter at 10 Hz.
2. The errors for the MEKF are in the order of $\approx 1 \times 10^{-3}$ degrees, which is an order of magnitude greater than the static attitude determination error. However, the MEKF is solving for an entire attitude trajectory along with an unknown gyro bias. The MEKF also includes more realistic sensor models that include biases, alignment errors, and random measurement noise from the sensors. The static attitude determination assumed all the measurement vectors were perfect which is one of the reasons the errors are an order of magnitude smaller.
3. The initial condition for this test was a random rotation with a standard deviation of 20 degrees was applied to the initial ground truth attitude. The bias was initialized as the ground truth bias plus a random vector sampled from with a standard deviation of 1×10^{-5} (which is the order of magnitude for the true bias value). Figures 18 - 21, show the element wise comparison between the estimated quaternion and the ground truth quaternion.

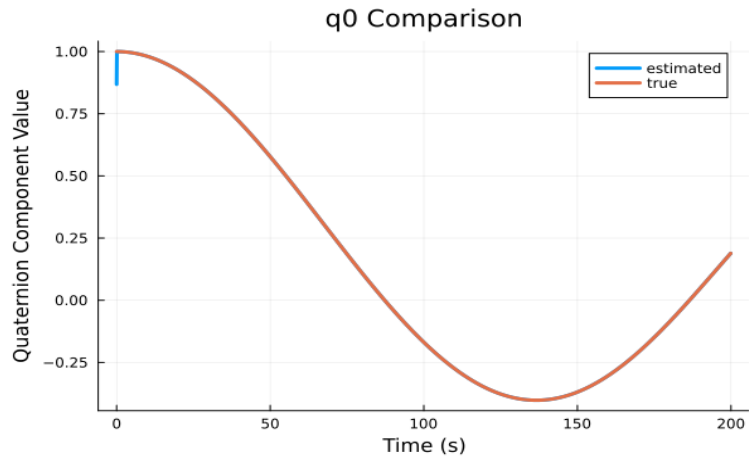


Figure 18: Scalar Part of the Quaternion True vs Estimated Plot

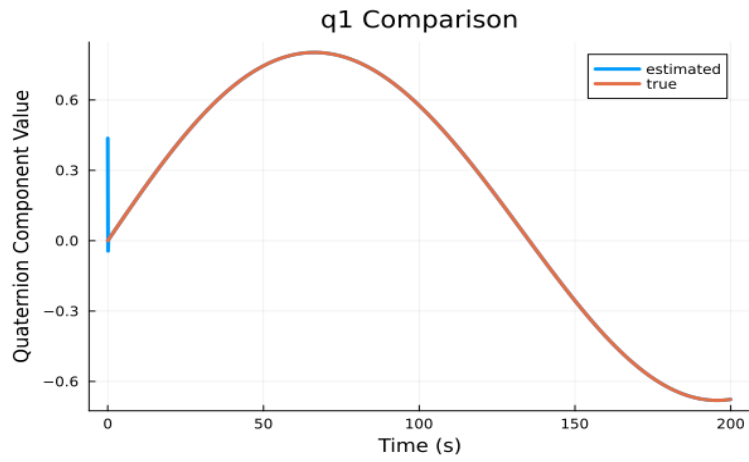


Figure 19: Vector Part of the Quaternion (x) True vs Estimated Plot

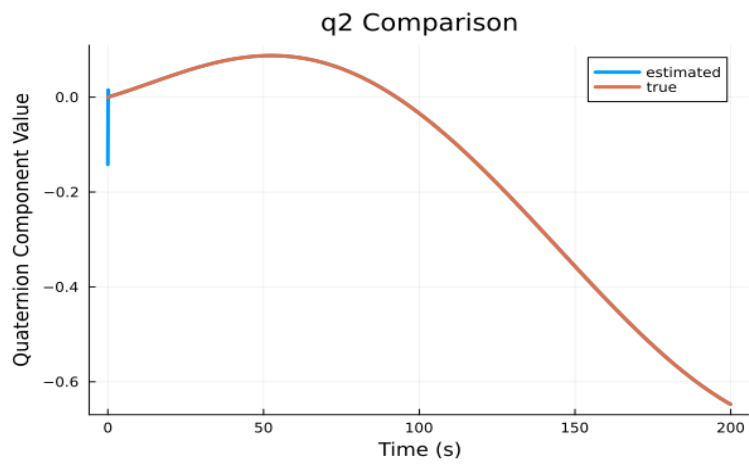


Figure 20: Scalar Part of the Quaternion (y) True vs Estimated Plot

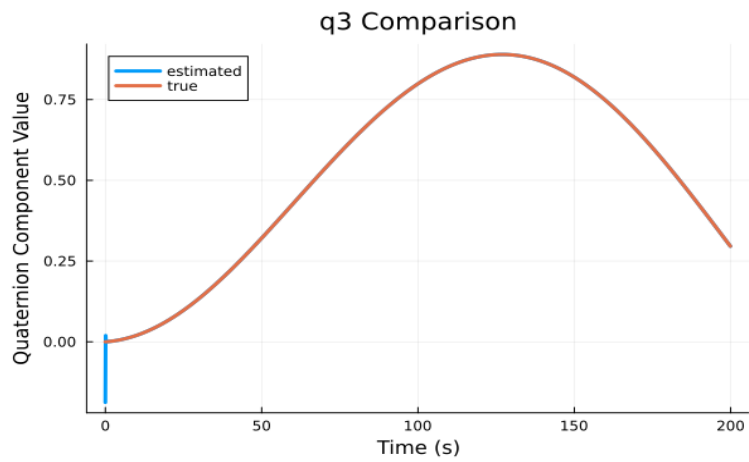


Figure 21: Scalar Part of the Quaternion (z) True vs Estimated Plot

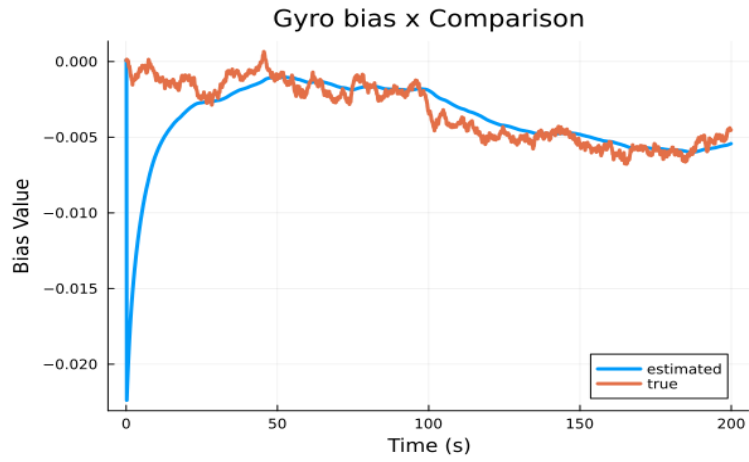


Figure 22: Gyro Bias x component True vs Estimated Plot

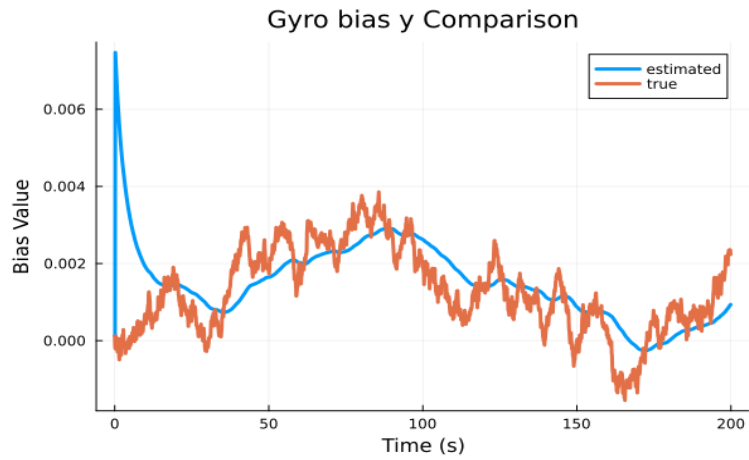


Figure 23: Gyro Bias y component True vs Estimated Plot

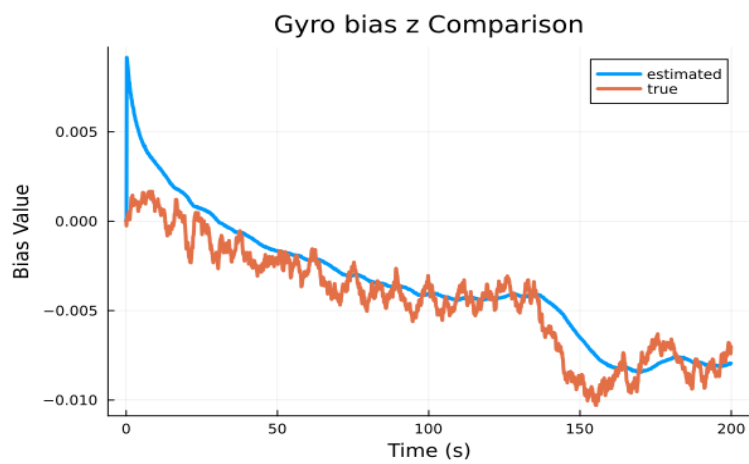


Figure 24: Gyro Bias z component True vs Estimated Plot

Next, we show the filter is consistent by plotting residuals between the ground truth and the estimated attitude and gyro bias. For a filter to be consistent, these residuals must exhibit the behavior of a zero mean Gaussian and bounded between the three-sigma covariance bounds that the filter estimates. This shows that the uncertainty estimate in the covariance is accurate and not over confident on the estimates. All the residual plots are start at 1 second for clarity. This is because in the beginning, there is a large residual due to the initial condition, however, it is consistent after a couple timesteps.

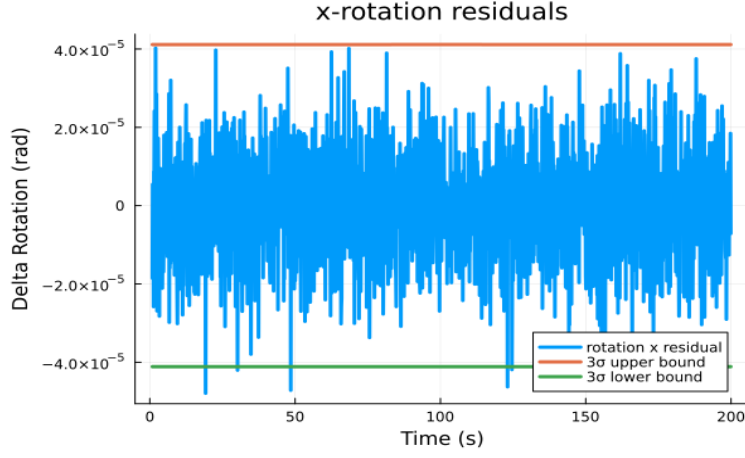


Figure 25: Rotation x component Residuals

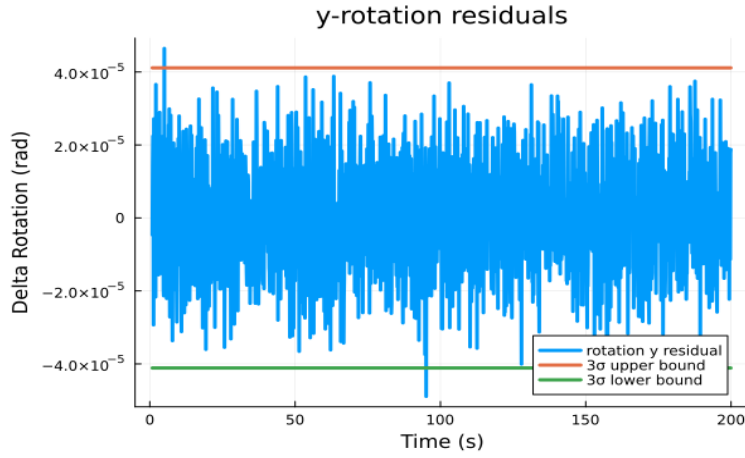


Figure 26: Rotation y component Residuals

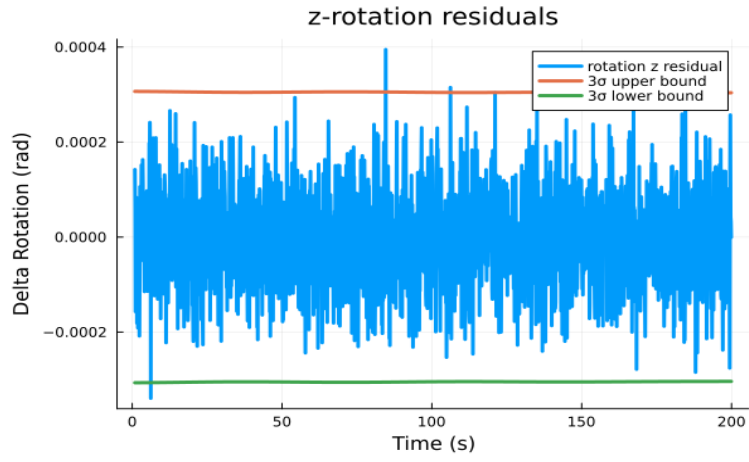


Figure 27: Rotation z component Residuals

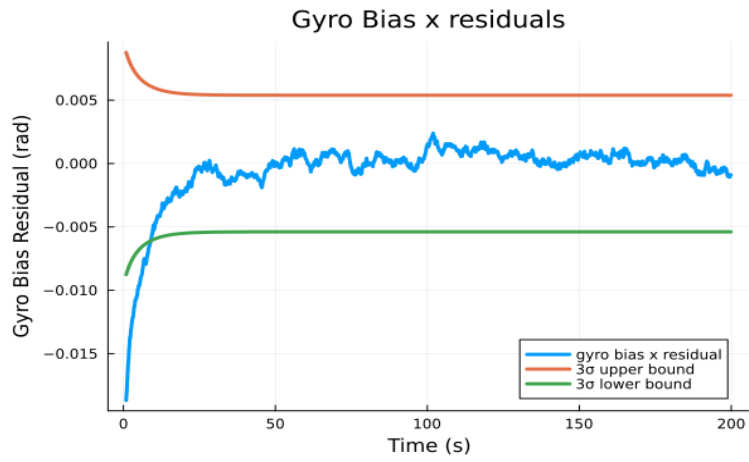


Figure 28: Gyro Bias x component Residuals

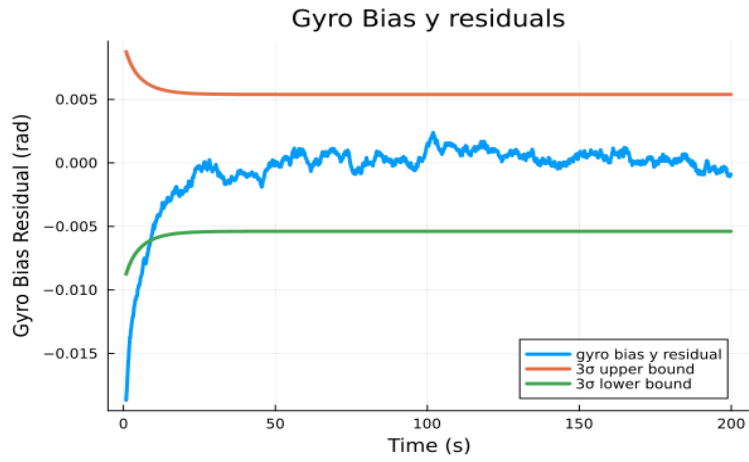


Figure 29: Gyro Bias y component Residuals

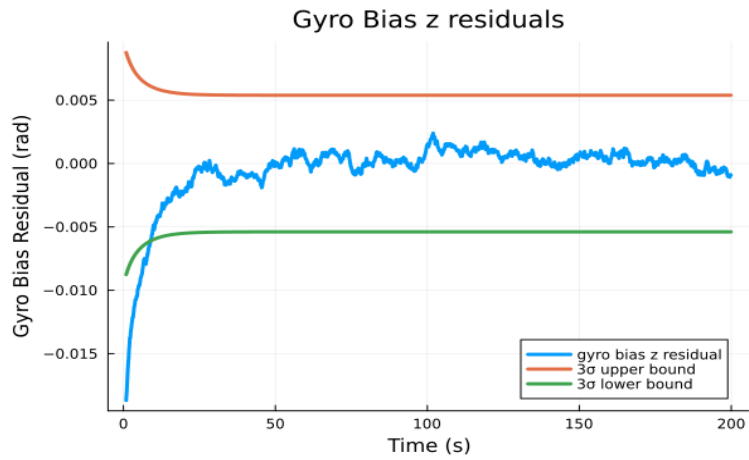


Figure 30: Gyro Bias z component Residuals

The 3-sigma bound on the gyro bias can be tighter and improved by tuning the process noise on the gyro bias.

4. For the test in Part 3, the attitude reduced down steady state after 3 iterations of the filter, which is 0.3 seconds, and the bias takes about 25 iterations to reach a steady state toward zero, which is equivalent to 2.5 seconds. Next we conduct an initial guess that is larger than the ground truth. We set the initial condition off by a random rotation with a 1-sigma standard deviation of 30 degrees and the gyro off by a random additive perturbation with a 1-sigma standard deviation of 1×10^{-4} . The filter is consistent and the residual plots are shown in Figures 31 - 36.

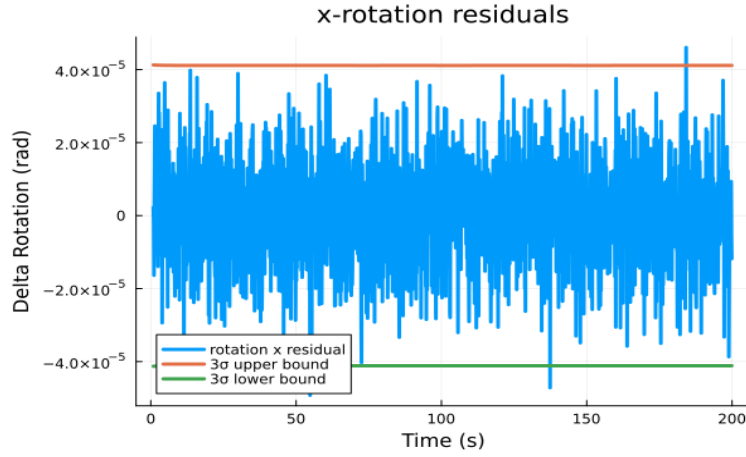


Figure 31: Rotation x component Residuals

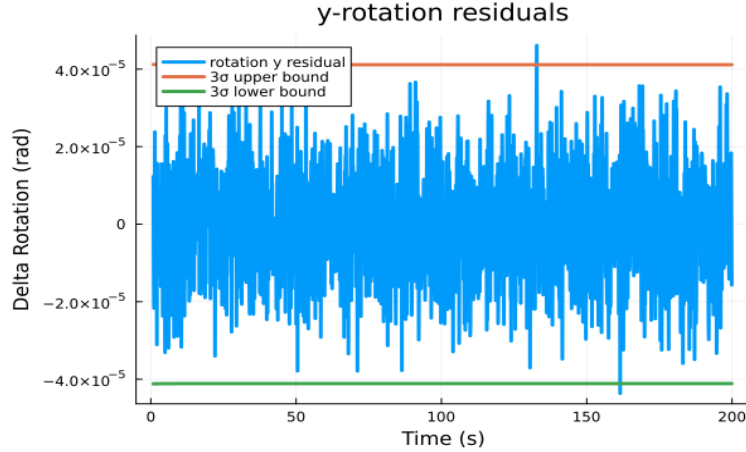


Figure 32: Rotation y component Residuals

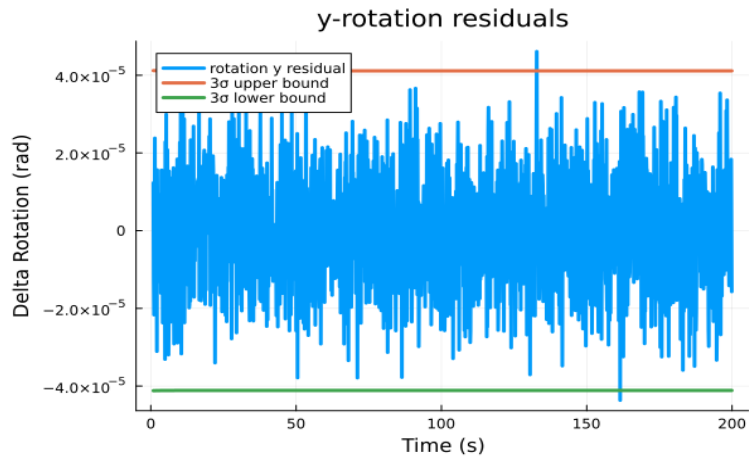


Figure 33: Rotation z component Residuals

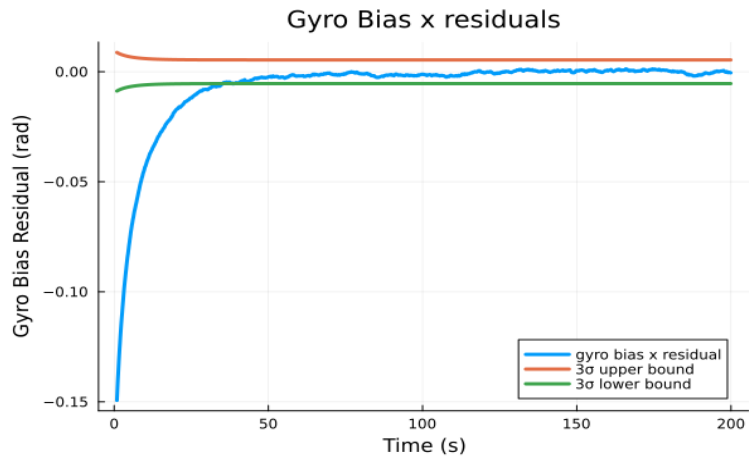


Figure 34: Gyro Bias x component Residuals

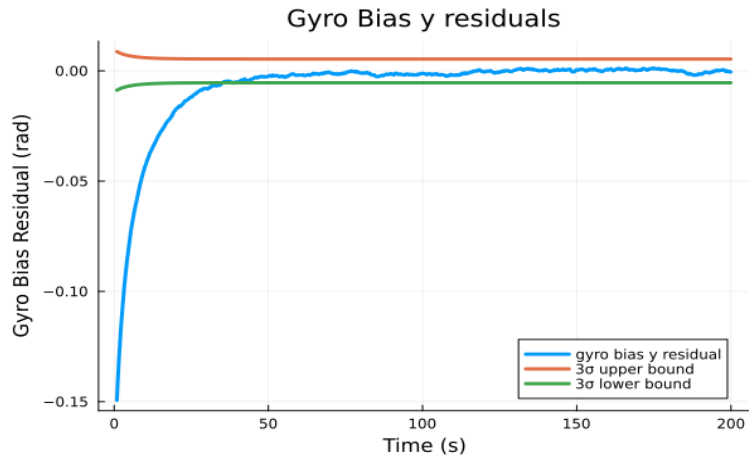


Figure 35: Gyro Bias y component Residuals

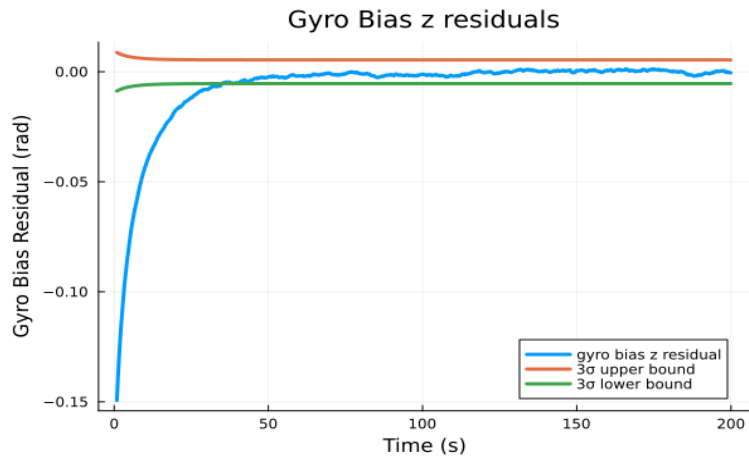


Figure 36: Gyro Bias z component Residuals

Next, we set the initial condition to a random rotation off the initial attitude with a 1-sigma standard deviation of 40 degrees and the gyro off by a random additive perturbation with a 1-sigma standard deviation of 1×10^{-4} . The filter is consistent and the residual plots are shown in Figures 37 - 42.

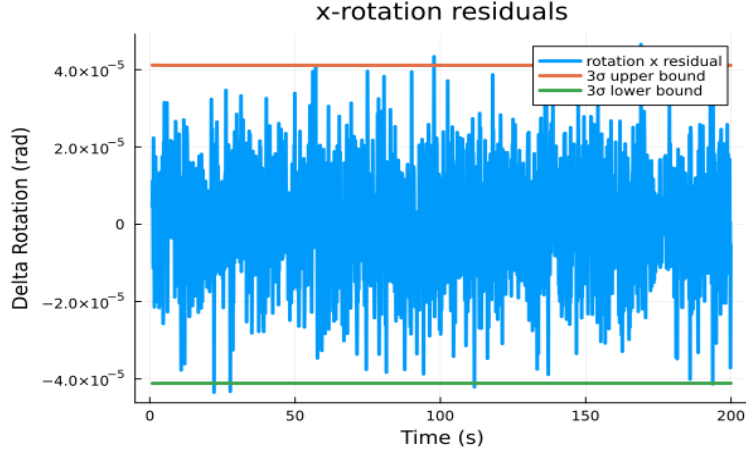


Figure 37: Rotation x component Residuals

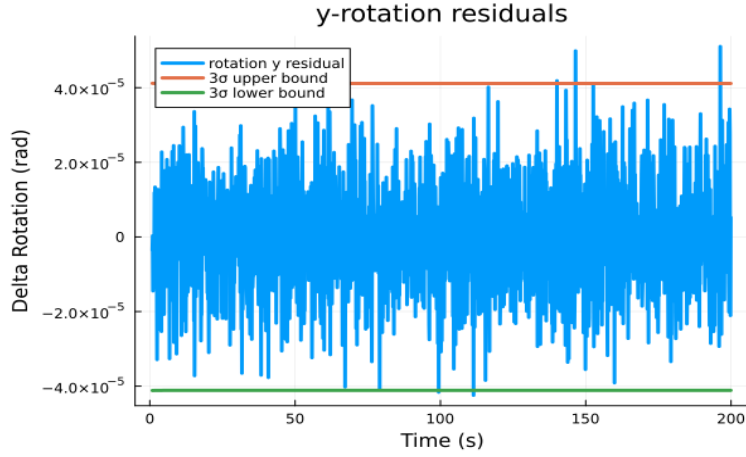


Figure 38: Rotation y component Residuals

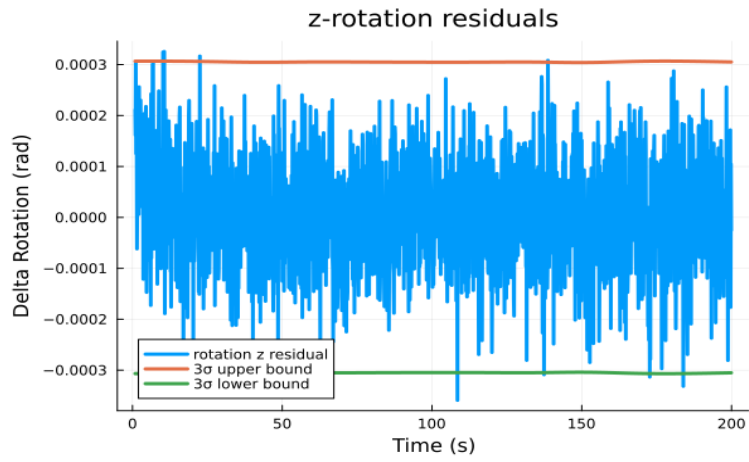


Figure 39: Rotation z component Residuals

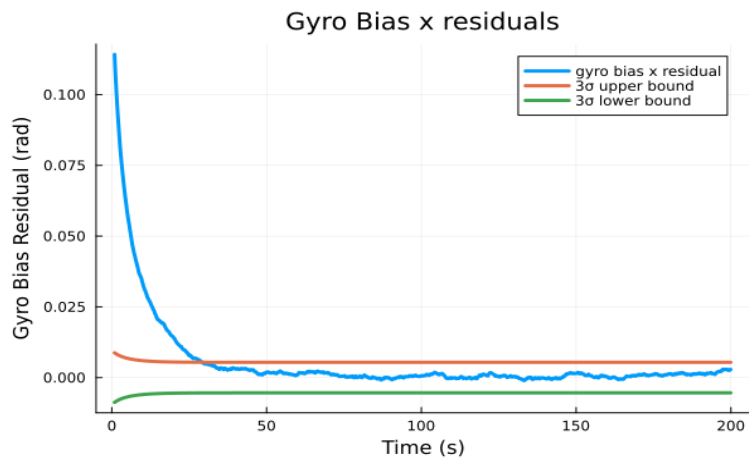


Figure 40: Gyro Bias x component Residuals

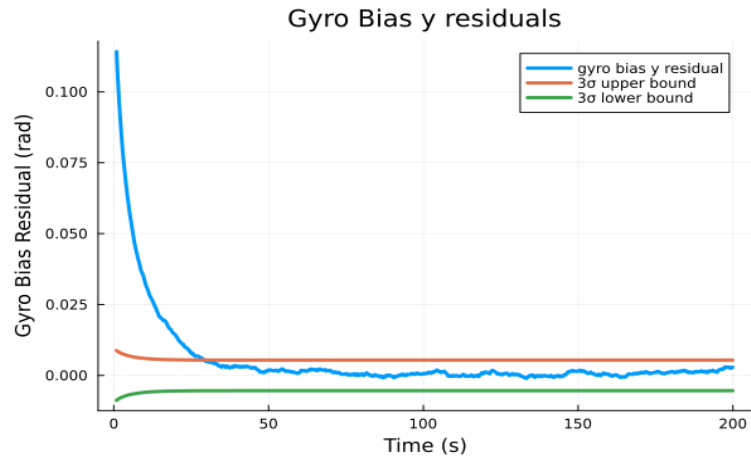


Figure 41: Gyro Bias y component Residuals

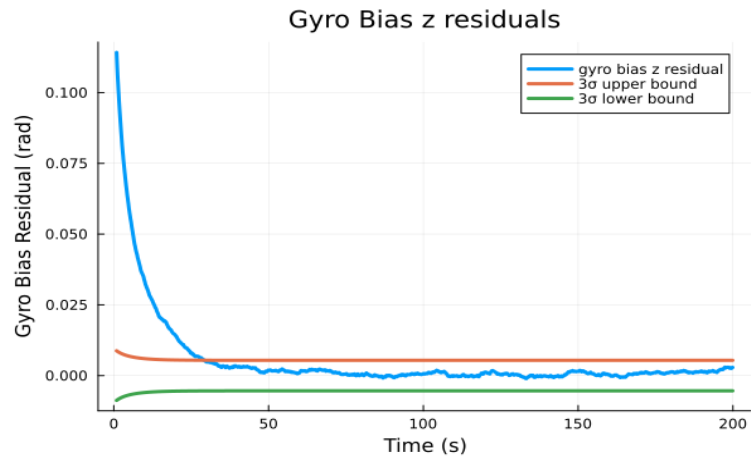


Figure 42: Gyro Bias z component Residuals

The filter is robust to large initial conditions (tested up to 40 degrees) as it converges in less than 1 second to steady state for the attitude and 5 seconds for the gyro bias. However, one observation is once the rotation error in the initial condition is high (> 50 degrees) compared to the ground truth, the double cover of quaternions is a problem as the residuals from the estimated and ground truth are zero mean, however, the quaternion comparison plots look completely different.

Actuator Specifications

1. The spacecraft has 3 reaction wheels providing a maximum momentum of 0.015 Nm s on each axis [4]. The propulsion system has 4 nozzles that each provide a minimum operating thrust of 4.2 mN and can provide a total of 30 m/s of delta V [3].
2. The four nozzles for the thrusters are pointed in the Z direction angled 20 degrees inward toward the X-Z plane [3]. We assume the reaction wheels are along each of the body axes.
3. The torque jacobian is calculated with the following expression

$$B_T = \begin{bmatrix} {}^B r_1 \times {}^B a_1 & {}^B r_2 \times {}^B a_2 & {}^B r_3 \times {}^B a_3 & {}^B r_4 \times {}^B a_4 \end{bmatrix} \quad (20)$$

$$B_T = \begin{bmatrix} -0.105 & -0.105 & 0.105 & 0.105 \\ 0.094 & -0.094 & 0.094 & -0.094 \\ -0.034 & 0.034 & 0.034 & -0.034 \end{bmatrix} \quad (21)$$

where ${}^B r_i$ is the thruster position relative to the COM and ${}^B a_i$ is the thruster axis unit vector. This torque jacobian multiplied by the thruster force commands provides the torque on the spacecraft in the body frame. The wheel jacobian is calculated using Eq. 22 where ${}^B a_i$ are the wheel spin-axis unit vectors. We assume that the spin axes are along the body frame vectors, therefore the wheel jacobian is the identity.

$$B_w = \begin{bmatrix} {}^B a_1 & {}^B a_2 & {}^B a_3 \end{bmatrix} \quad (22)$$

$$B_w = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (23)$$

Environmental Torques

1. The aerodynamic drag torque on a surface element with an outward normal $\hat{N}$ on a surface element dA is given by

$$df_{aero} = -\frac{1}{2} C_D \rho V^2 (\hat{N} \cdot \hat{V}) \hat{V} dA \quad (24)$$

where $\hat{V}$ is the unit vector in the direction of translational velocity V of the surface element relative to the incident stream, ρ is the atmospheric density

(calculated with the Harris Priester model in this implementation). The aerodynamic torque τ_{aero} due to the force df_{aero} is

$$\tau_{aero} = \int r_s \times df_{aero} \quad (25)$$

where r_s is the distance between the center of mass of the face to the center of mass of the spacecraft. This integral is over the spacecraft surfaces in which $\hat{N} \cdot \hat{V} > 0$ as this will create a drag force in the negative direction of the velocity. The velocity in ECI relative to the atmosphere is calculated from [12]

$$V_0 = V_{ECI} - \omega_{EARTH} \times r \quad (26)$$

The translational velocity of element dA for a spacecraft is

$$V = V_0 + \omega_{rel} \times r_s \quad (27)$$

where V_0 is the velocity relative to the atmosphere, ω_{rel} is the angular velocity relative to the rotation of the atmosphere (which approximately equals the Earth rotation rate) [13]. The total aerodynamic force is calculated by summing up the force contributions over the individual satellite faces. The total torque about the center of mass is the sum of the individual torques calculated by the cross product of the distance from the spacecraft center of mass to the center of pressure of each of the faces (we assume the center of mass is the center of pressure). The gravity gradient torque (τ_{gg}) is calculated by

$$\tau_{gg} = \frac{3\mu}{(r^T r)^{\frac{5}{2}}} r \times Jr \quad (28)$$

where r is the position of the spacecraft in ECI, J is the body inertia, and μ is the Earth gravitational parameter.

Next, we simulate the dynamics with and without these perturbations and list the maximum magnitude of each perturbation, along with their average and the change in angular momentum. A drag acceleration was also implemented in the dynamics model and the result is shown in Fig. 43. Over 10 revolutions, the drag acceleration perturbation caused a change of 2.14 km compared to the orbit simulated without the drag perturbation (Fig. 3). Figures 44 and 45 show the gravity gradient torque and drag torque over a 10 revolution period. The maximum magnitude of the gravity gradient torque along ten orbits was 7.584×10^{-7} Nm and the maximum magnitude of the drag torque is 1.217×10^{-7} Nm.

The drag and gravity gradient perturbations have an effect on the angular velocity and attitude trajectory of the spacecraft. This difference is shown in Figures 46 and 47. The initial angular velocity was randomly sampled from a normal distribution with a standard deviation of 2 degrees per second.

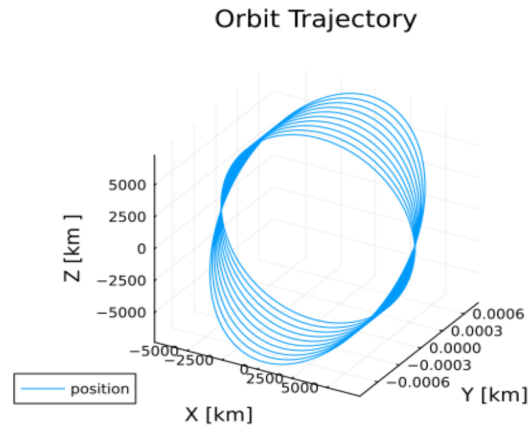


Figure 43: Orbit Trajectory with Drag

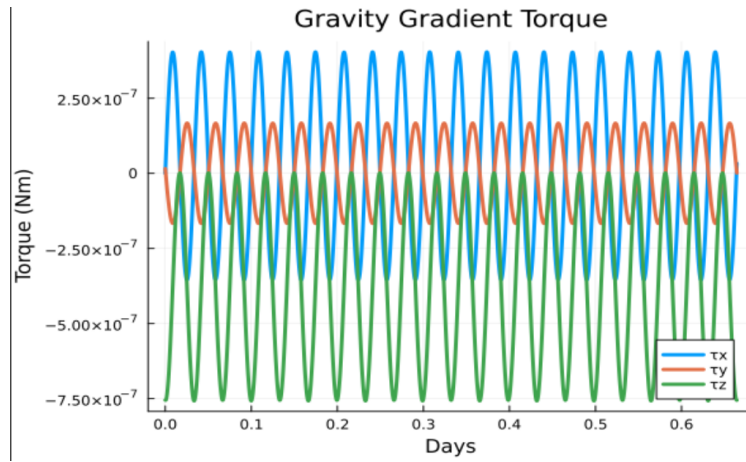


Figure 44: Gravity Gradient Torque

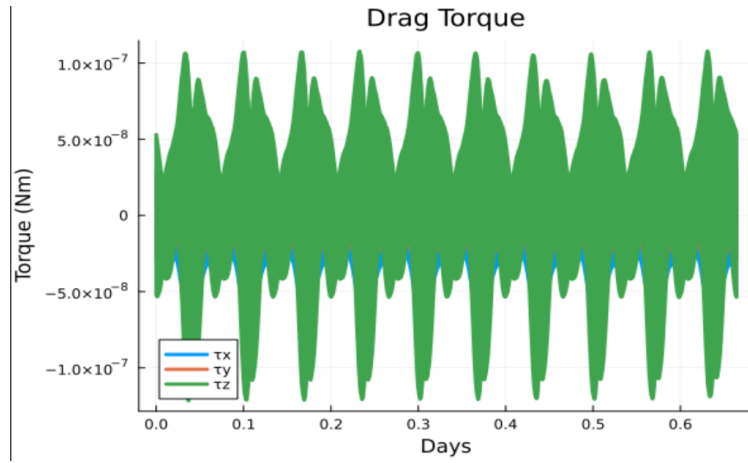


Figure 45: Drag Torque

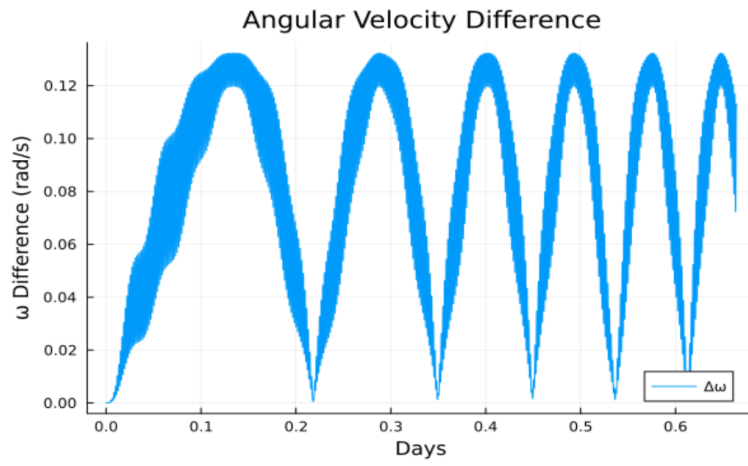


Figure 46: Difference in angular velocity between the dynamics with environmental perturbations and no perturbations

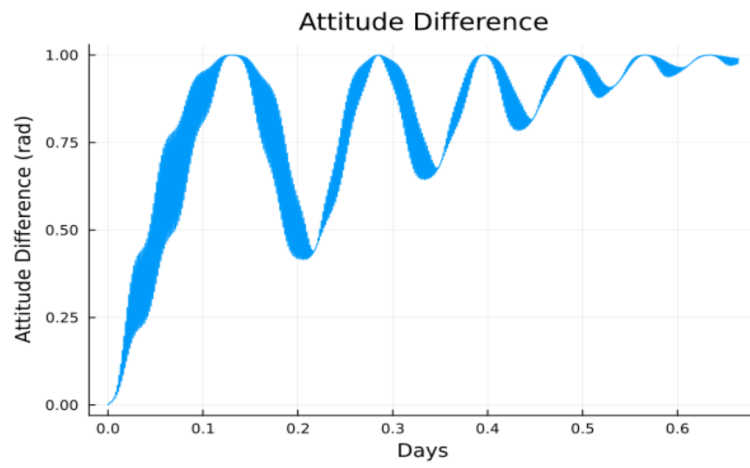


Figure 47: Difference in attitude between the dynamics with environmental perturbations and no perturbations

2. The average drag torque over one orbit is

$$\bar{\tau} = \begin{bmatrix} 1.395 \times 10^{-10} \\ -1.374 \times 10^{-10} \\ -1.080 \times 10^{-9} \end{bmatrix} \text{ Nm}$$

and the average gravity gradient torque over one orbit is

$$\bar{\tau} = \begin{bmatrix} -6.175 \times 10^{-8} \\ -6.405 \times 10^{-12} \\ -3.058 \times 10^{-7} \end{bmatrix} \text{ Nm}$$

The maximum accumulated change in angular momentum expected over a full day is

$$h = 0.0405 \text{ Nms}$$

3. The maximum accumulated change in angular momentum is greater than the maximum momentum of the reaction wheels (0.015 Nms in each axis). If this limit is reached, then the reaction wheels will saturate and not be able to provide the torque needed for attitude control. Therefore, momentum dumping would have to be performed once every 5 orbits (approximately every 8 hours) to prevent it from reaching the hardware limit.

Attitude Regulation

An LQR controller is implemented to maintain the desired attitude. We first obtain the cost to go matrices P and linear feedback control matrices K using a backward Ricatti recursion. Since this problem is a regulation problem, the gain matrix K from the Ricatti recursion, converges to a steady state value. We use this infinite horizon constant K throughout the entire attitude regulation sequence. Initially we sample a desired attitude from in the range ± 90 degree away from the ground truth attitude. We also initialize our multiplicative extended Kalman filter to be sampled from a Gaussian with a ± 20 degrees 1 sigma covariance away from the ground truth. The Q and R matrices were used to ensure the controls did not violate the thruster limits or reaction wheel torque limits (by increasing R). The estimated MEKF state is used to calculate the error state. Once the error state is obtained, the control law is the gain matrix multiplied by the error state. Then, this control is used to simulate the ground truth simulation 1 step forward and then measurements are gathered at this new ground truth state. These measurements are the used as inputs for the MEKF to predict the next state at timestep $k + 1$. These set of steps repeat

until the end of the simulation. The linear system used in the LQR controller is the following:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\omega} \\ \dot{\rho} \end{bmatrix} = \begin{bmatrix} 0 & I & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \omega \\ \rho \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -J^{-1}B_w & J^{-1}B_T \\ B_w & 0 \end{bmatrix} \begin{bmatrix} w \\ T_F \end{bmatrix} \quad (29)$$

This system is linearized about the desired attitude quaternion, zero angular velocity, and zero wheel momentum. The state is the axis angle error state, angular velocity, and wheel momentum. The control is the wheel torque and the thruster forces. The wheel torque and thruster forces are mapped into body torques using their respective actuator jacobians B_w and B_T (derived in the actuator specifications section).

We simulate the controller with an MEKF estimate and noisy measurements. We perform two tests with random desired attitudes ± 90 degrees away from the initial attitude and show the attitude converges to desired attitude while respecting attitude limits, and the angular velocity and wheel momentum terms also converge to zero. The performance is summarized in Figures 48 - 67.

1.

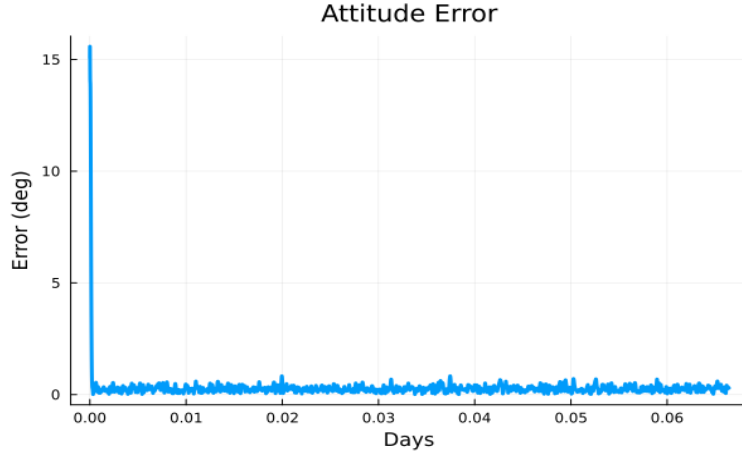


Figure 48: Attitude Error Test 1

2. After the correction maneuver to the desired attitude, the RMSE pointing error is 0.122 degrees.

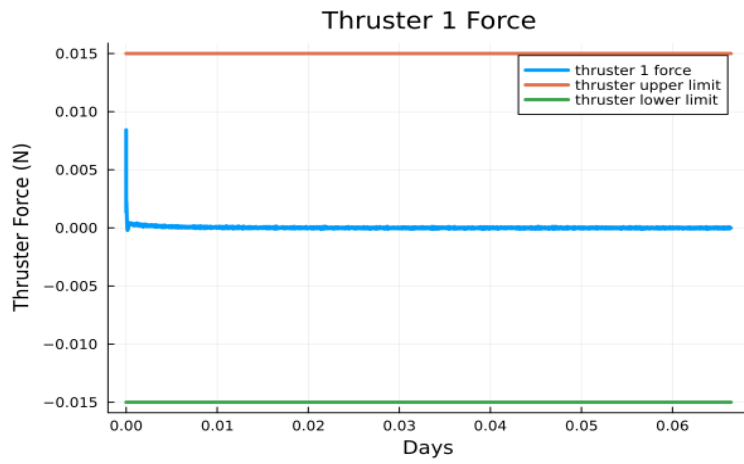


Figure 49: Force from Thruster 1 Test 1

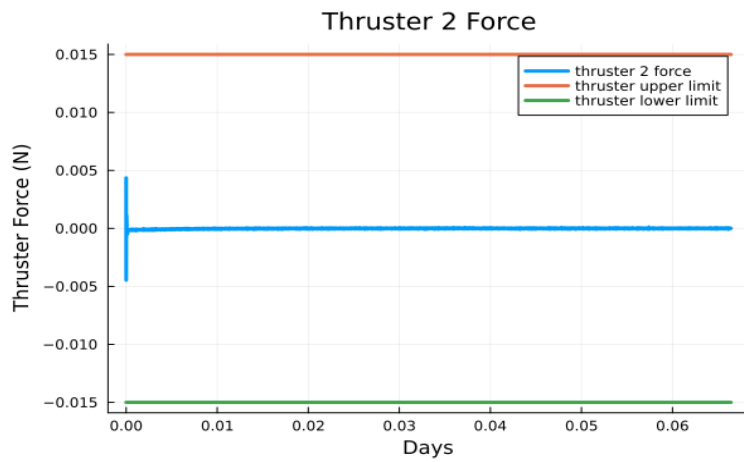


Figure 50: Force from Thruster 2 Test 1

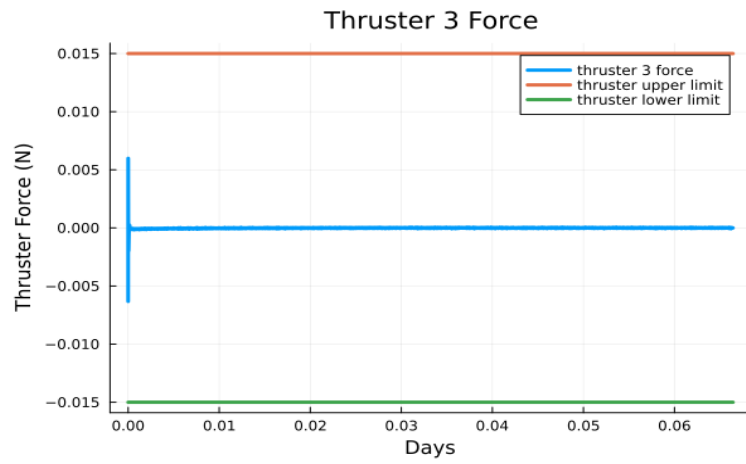


Figure 51: Force from Thruster 3 Test 1

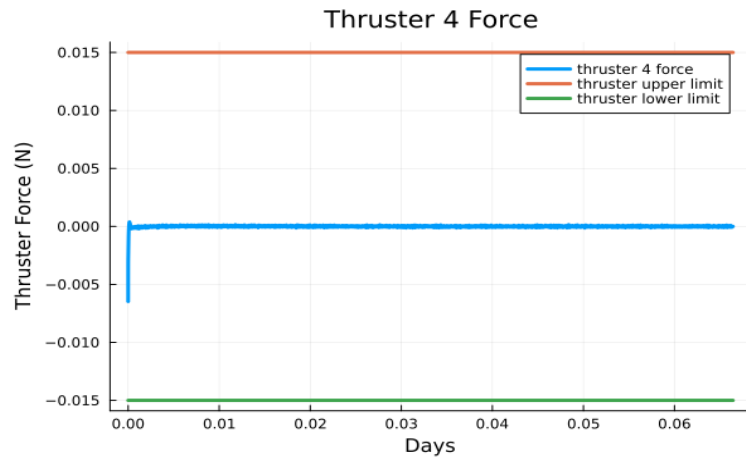


Figure 52: Force from Thruster 4 Test 1

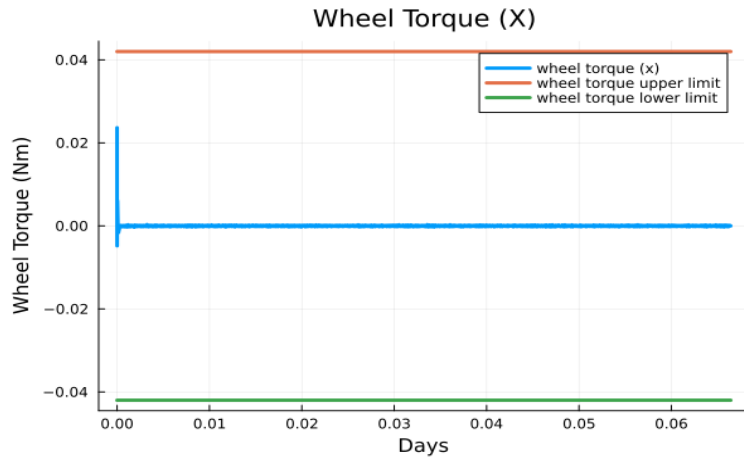


Figure 53: Wheel Torque X Test 1

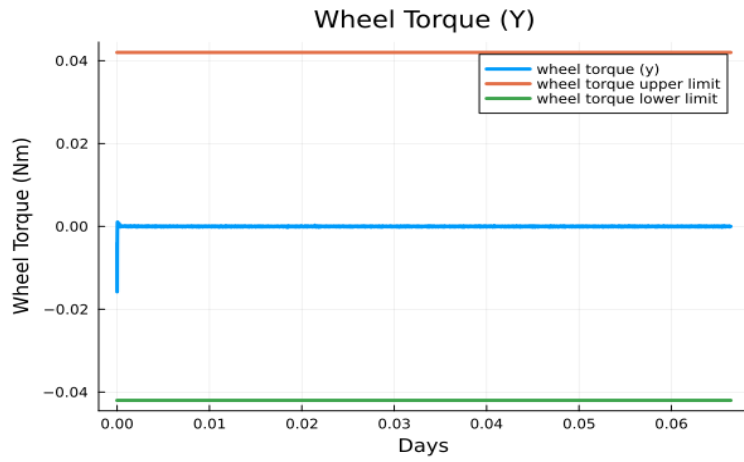


Figure 54: Wheel Torque Y Test 1

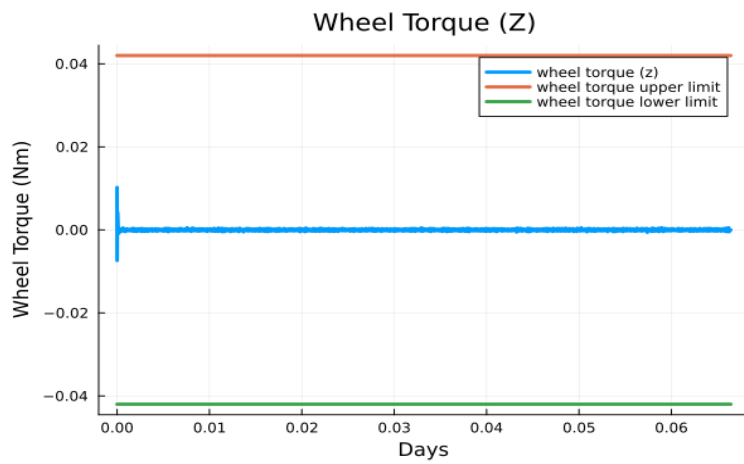


Figure 55: Wheel Torque Z Test 1

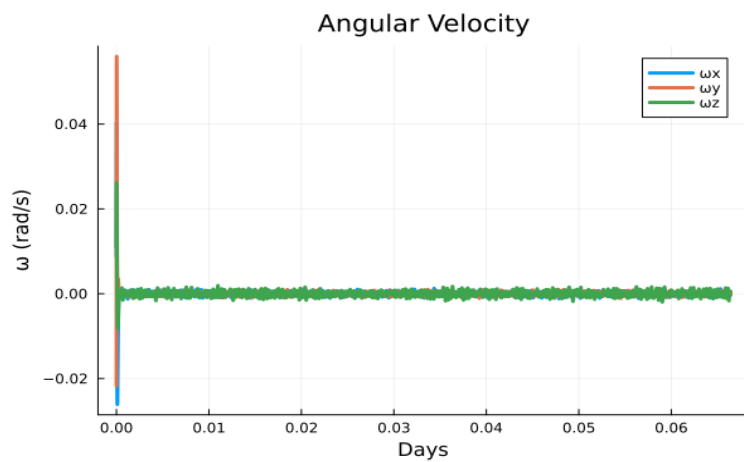


Figure 56: Angular Velocity Test 1

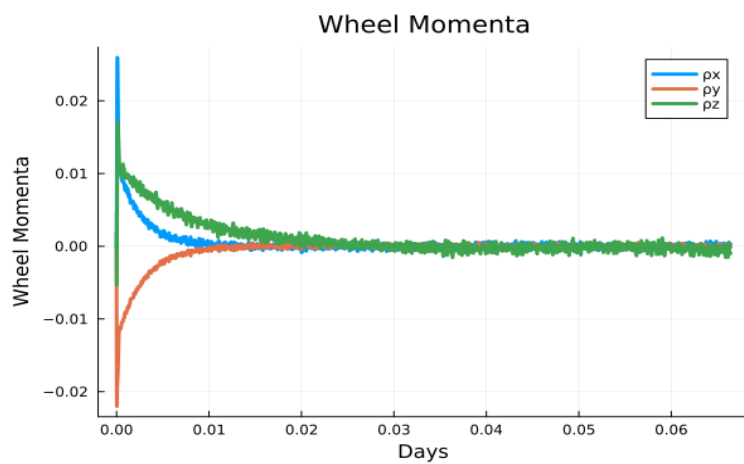


Figure 57: Wheel Momenta Test 1

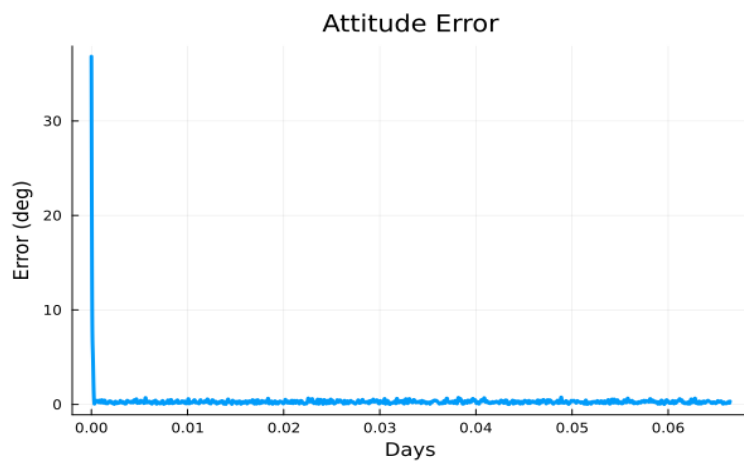


Figure 58: Attitude Error Test 2

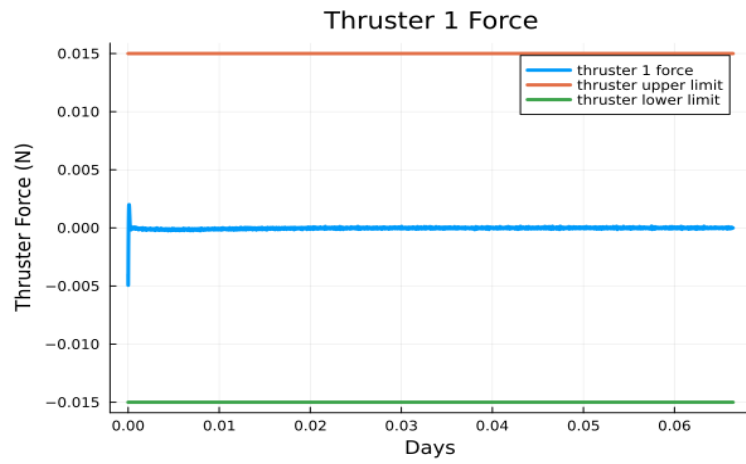


Figure 59: Force from Thruster 1 Test 2

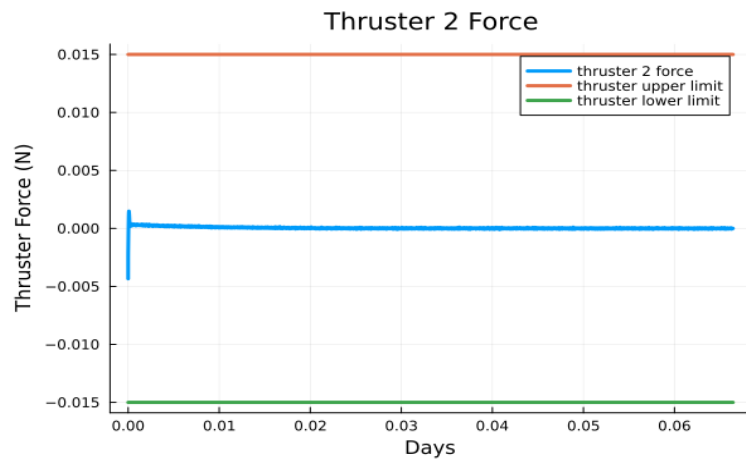


Figure 60: Force from Thruster 2 Test 2

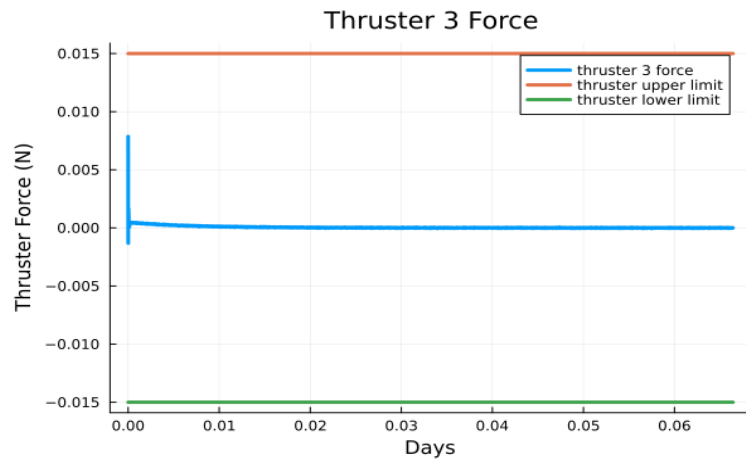


Figure 61: Force from Thruster 3 Test 2

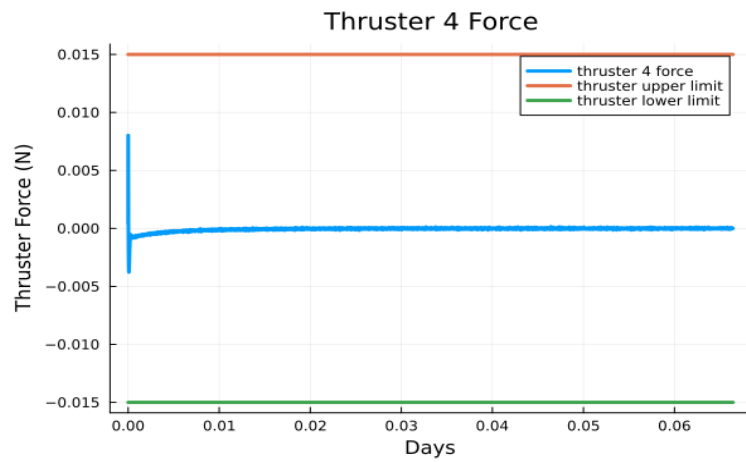


Figure 62: Force from Thruster 4 Test 2

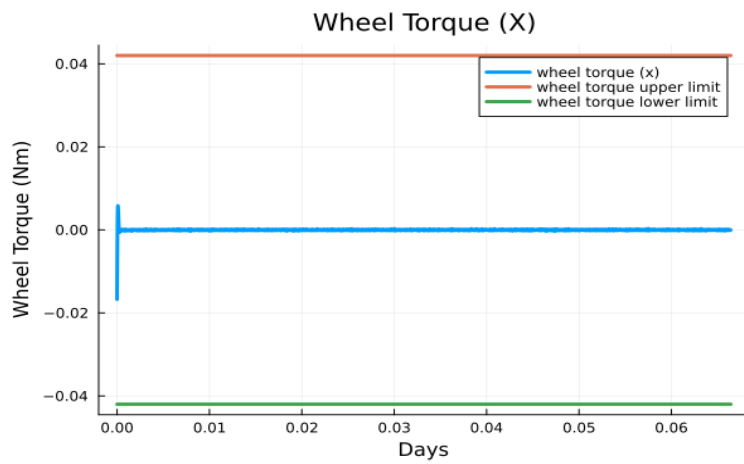


Figure 63: Wheel Torque X Test 2

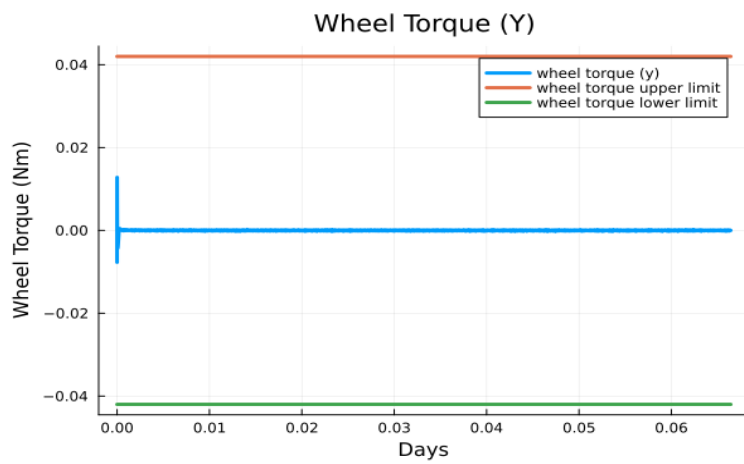


Figure 64: Wheel Torque Y Test 2

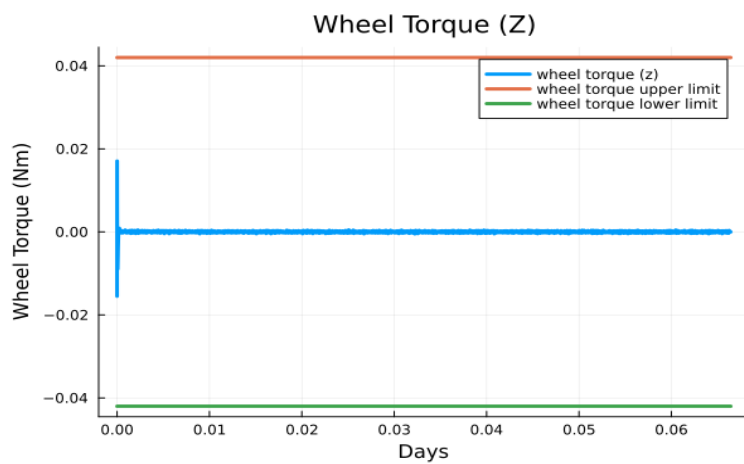


Figure 65: Wheel Torque Z Test 2

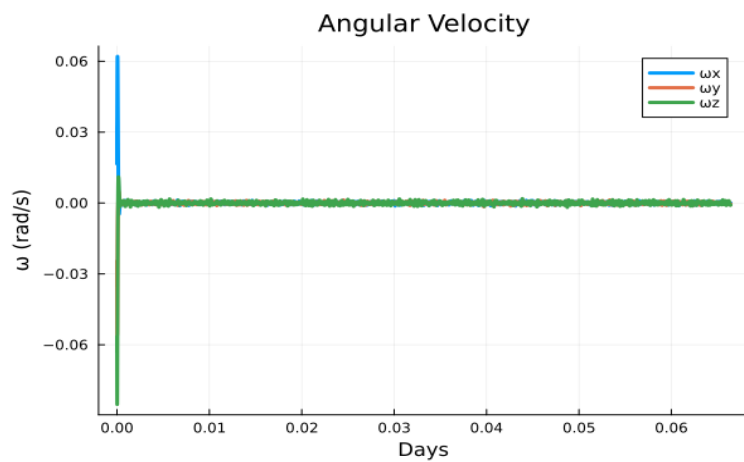


Figure 66: Angular Velocity Test 2

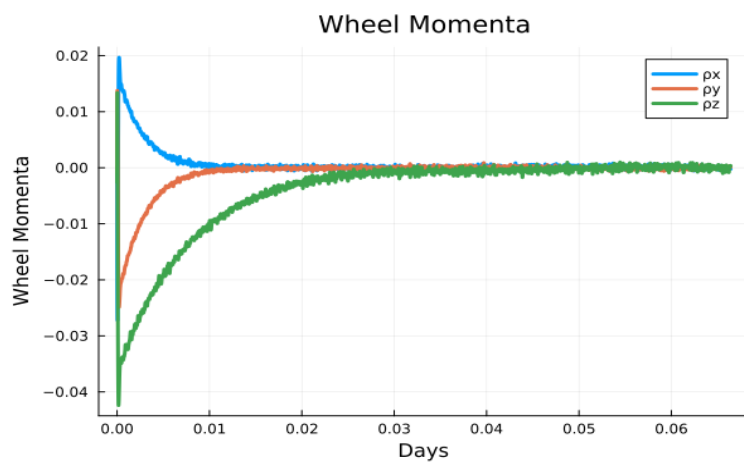


Figure 67: Wheel Momenta Test 2

Eigen-Axis Slew

For the eigen axis slew, we calculate the axis angle vector between the current attitude and the final attitude (using the quaternion log). The normalized vector of the axis angle is the axis of rotation and the norm is the angle needed along that axis to match both rotations. Using this axis angle, we then create a reference trajectory of attitude was generated using the following versine function

$$\theta(t) = \|\theta_{final}\| \frac{1}{2}(1 - \cos(\alpha t)) \quad (30)$$

where t is time, θ is the angle along the rotation axis, θ_{final} is the final angle at the end of the trajectory. We can also differentiate this trajectory to get $\omega(t)$ and $\dot{\omega}$. Next, we use inverse dynamics to solve for the controls (wheel torque $\dot{\rho}$), and can integrate the wheel torque to get the wheel momentum ρ . This provides an open-loop control trajectory corresponding to the reference trajectory from the versine function. We then implement a TVLQR controller to compensate model error and reject disturbances. We linearize the dynamics along the reference trajectory from the versine function. The linearized continuous dynamics system are

$$\begin{bmatrix} \dot{\phi} \\ \Delta \dot{\omega} \\ \Delta \dot{\rho} \end{bmatrix} = \begin{bmatrix} -\hat{\omega} & I & 0 \\ 0 & -J^{-1}[\hat{\omega}J - (J\omega + \hat{B}_w\rho)] & -J^{-1}\hat{\omega}B_w \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \phi \\ \Delta \omega \\ \Delta \rho \end{bmatrix} + \begin{bmatrix} 0 \\ -J^{-1}B_w \\ B_w \end{bmatrix} u \quad (31)$$

These dynamics are then discretized using the matrix exponential and used in the Ricatti recursion to obtain a linear feedback matrix K at each timestep along the trajectory. The total time of the maneuver was set to 300 seconds, and it is a 180 degree slew maneuver starting at the identity quaternion. We simulate the controller with an MEKF estimate and noisy measurements and the performance is summarized in Figures 68 - 73. Figures 73 and 68 show the TVLQR controller track the nominal trajectory well throughout the trajectory. There are large errors in the beginning, but this can be due to the MEKF uncertainty at the beginning of the simulation.

1.

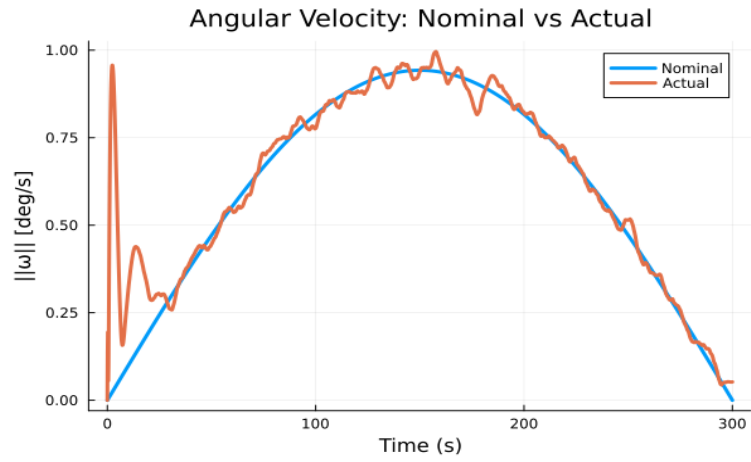


Figure 68: Actual vs Nominal Angular Velocity

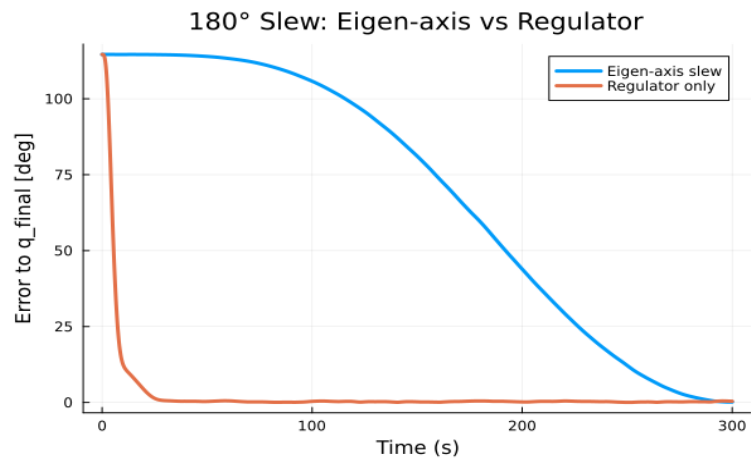


Figure 69: Eigen Axis Slew vs Regulator controller

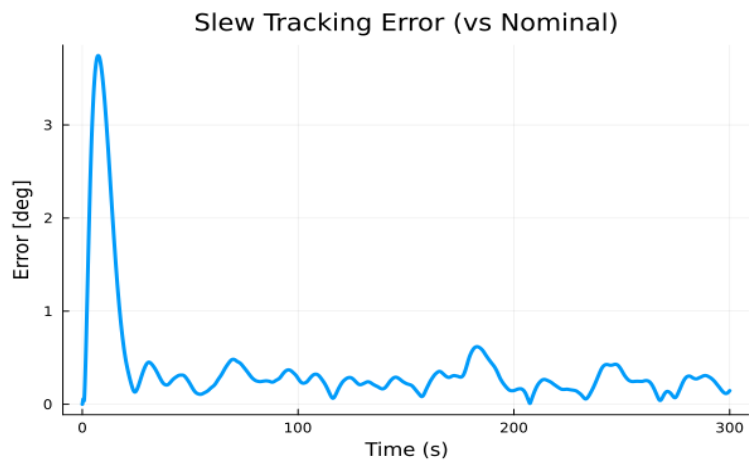


Figure 70: Tracking Error Eigne Axis Slew

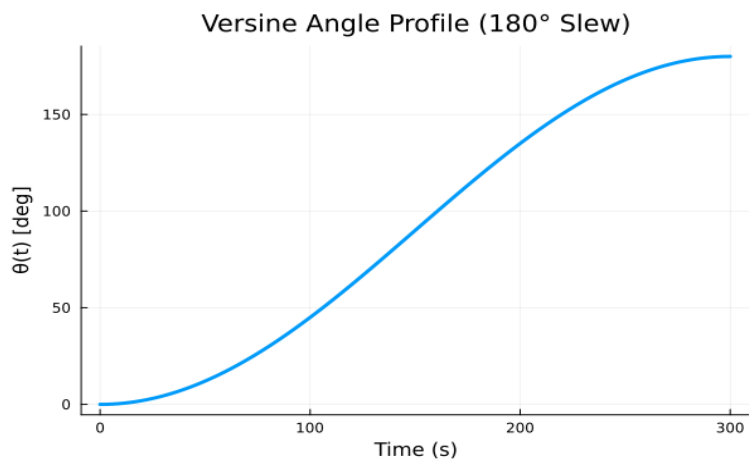


Figure 71: Versine Attitude Trajectory

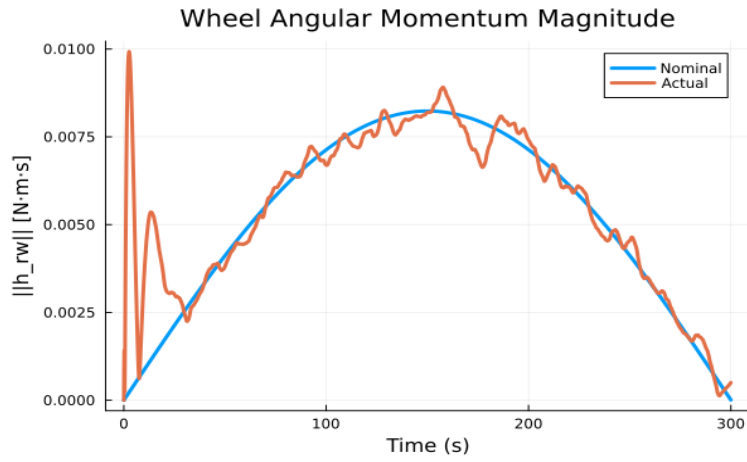


Figure 72: Wheel Momenta Eigen Axis Slew

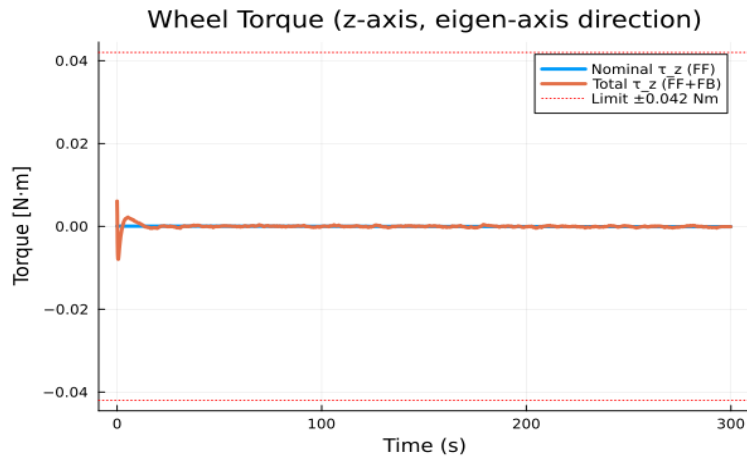


Figure 73: Wheel Torque Z Eigen Axis

2. Both the eigen-axis slew trajectory and the regulator trajectory reach the desired goal. However, the LQR slew trajectory saturates the controls and attempts to reach the trajectory as fast as possible. This can be managed by tuning Q and R , however, tuning a Q and R depending on the initial condition is not ideal. On the other hand, the open loop control trajectory in the eigen axis slew is the majority of the control used and the LQR controller is only correcting the environmental disturbances when simulating on the nonlinear dynamics. The controls stay within the bounds for the eigen axis slew and the angular velocity and wheel momentum track the reference trajectory well.

Control Barrier Functions for Control Moment Gyroscopes

Introduction

Control moment gyros (CMGs) are spacecraft actuators that exchange momentum to effectively reorient the spacecraft's attitude. They consist of a set of rotor axes that are allowed to rotate in the spacecraft body frame and each rotor includes a flywheel that spin at a constant speed. CMG's have an advantage over reaction wheels because of their torque amplification property that allows small gimbal motors to produce very large output torques, while reaction wheels are limited in torque capability because of mechanical power [14]. The arrangement of the CMG's on the spacecraft create different momentum envelopes that the spacecraft can leverage to maximize attitude tracking performance. Control of CMG's is typically decoupled into a torque controller and a steering law that transforms torques into gimbal rates. To avoid singularities, there is usually a redundant amount of CMG's on the spacecraft (at least 4) because the null space can be used to avoid singularities. However, operating near singularities can result in high gimbal rate commands and undesirable system response if the desired torque direction is in the singular direction. Many of the steering laws in the literature avoid singularities by introducing torque error. This torque error is introduced by regularizing the actuator jacobian when calculating the Moore-Penrose pseudoinverse. This alleviates ill-conditioning of the actuator jacobian, at the expense of not tracking the desired torque which may affect attitude tracking performance. Some steering law examples include the singularity-robust inverse [15] that uses a singularity measure as an additive parameter to regularize the Moore-Penrose pseudoinverse, a singularity direction avoidance psuedoinverse which only regularizes the singularity direction calculated from an SVD on the actuator jacobian [16], and a generalized singularity robust psuedoinverse in which the regularizer is parametrized by trigonometric functions [17]. All the pseudoinverse based control laws are dependent on a vast amount of parameters that do not guarantee singularity avoidance and add torque error to the control which affects attitude tracking performance. In this report, we explore the use of control barrier functions as a formal guarantee to avoid CMG singularities while targeting a desired attitude effectively. We then compare our formulation to a regularized Moore Penrose psuedoinverse solution.

Method

We initialize the CMG's in a box 90 configuration as shown in Fig. 74. The state x consists of the spacecraft quaternion q , angular velocity ω , and the axes of rotation for each CMG $a_i \forall i = [1, \dots, 4]$; the continuous dynamics are shown in Eq. 32. The term $B_w(x)$ is the CMG actuator jacobian, J is the spacecraft moment of inertia in the body frame, $G(q)$ is the attitude jacobian, w are the constant wheel speeds, and g are the fixed gimbal axes in the spacecraft body

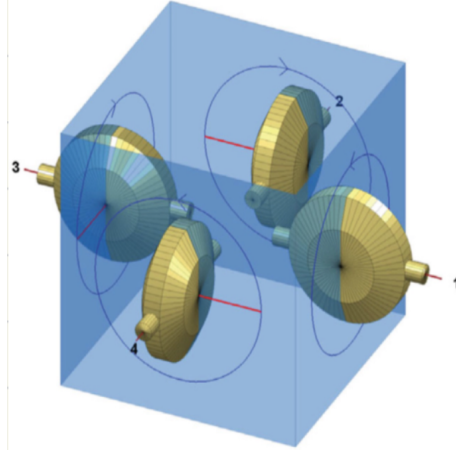


Figure 74: Box 90 CMG Array Configuration

frame.

$$\begin{bmatrix} \dot{q} \\ \dot{\omega} \\ \dot{a}_1 \\ \dot{a}_2 \\ \dot{a}_3 \\ \dot{a}_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}G(q)\omega \\ -J^{-1}[\omega \times (J\omega + \sum_{i=1}^4 w_i a_i) - B_w(x)u] \\ (g_1 \times a_1)u_1 \\ (g_1 \times a_2)u_2 \\ (g_1 \times a_3)u_3 \\ (g_1 \times a_4)u_4 \end{bmatrix} \quad (32)$$

A fourth-order Runge-Kutta integrator is then used to integrate these dynamics forward in time. A desired torque to achieve a reference attitude is calculated using a PD controller with proportional gain k_p and derivative gain k_d which multiply by the attitude error state (expressed as an axis-angle ϕ) and the angular velocity ω deviation from zero as shown in Eq. 33.

$$\tau_d = -k_p \phi - k_d \omega \quad (33)$$

The CMG actuator jacobian maps four gimbal rates into three output torques. This system is undetermined as there is a family of solutions that can produce the three dimensional torque because vectors in the null space of the actuator jacobian can be added to the gimbal rate vector and not change the output torque but have an effect on the gimbal rates. Specifically, if u^* tracks the desired torque and satisfies $B_w(x)u^* = \tau$, then $u^* + v \cdot n_b$ is also a valid solution where n_b is the null space vector of $B_w(x)$ and v a positive scalar value. The CMG actuator jacobian is defined in Eq. 34

$$B_w(x) = \begin{bmatrix} -w_1(g_1 \times a_1) & -w_2(g_2 \times a_2) & -w_3(g_3 \times a_3) & -w_4(g_4 \times a_4) \end{bmatrix} \quad (34)$$

Singularity avoidance can be posed as a set invariance problem. Control barrier functions define a safety set and leverage optimization-based controllers to ensure invariance in the safe set throughout the trajectory [18]. To do this,

we define the control barrier function $h(x)$ as the manipulability index [19], a common metric in robotic manipulators that approximates the distance to a singularity. Eq. 35 shows the control barrier function and ε is a user defined tolerance that sets the lower bound to the control barrier function.

$$h(x) = \sqrt{\det(B_w(x)B_w(x)^T)} - \varepsilon \quad (35)$$

The goal is to ensure $h(x) > 0$ (forward invariance) along the entire control trajectory, as it ensures that the CMG arrays do not approach a singular configuration. This can be achieved with a controller satisfying Eq. 36,

$$\dot{h}(x, u) \geq -\alpha(h(x)) \quad (36)$$

where $\dot{h}(x, u)$ is the time derivative of the control barrier function (an affine function in u), and α is a K_∞ function (set to linear) [20]. By the chain rule $\dot{h} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t}$, where the second term are the dynamics. Since $B_w(x)$ depends only on the axes a , $\dot{x}$ is equivalent to $\dot{a}$ in Eq. 32 which is affine in u . Then, we incorporate the condition in Eq. 36 into a quadratic program which is a convex optimization problem that can be solved to global optimality efficiently using off-the-shelf solvers [21]. The quadratic program is formulated in Eq. 37,

$$\begin{aligned} \min_{u \in \mathbb{R}^4, s \in \mathbb{R}^3} \quad & \beta \|s\|^2 + \lambda (n_B^\top u)^2 \\ \text{subject to} \quad & \dot{h}(x, u) + \alpha h(x) \geq 0, \\ & B(x)u + s = \tau_d, \end{aligned} \quad (37)$$

where $u \in \mathbb{R}^4$ is the control input, $s \in \mathbb{R}^3$ is a slack variable penalizing torque error, $B(x) \in \mathbb{R}^{3 \times 4}$, τ_d is the desired torque target, $h(x)$ is the CBF value, $\dot{h}$ is the CBF derivative, n_B is the null space vector of the actuator jacobian $B_w(x)$, and $\lambda = 1$, $\alpha = 2.0$, $\beta = 1000$. This formulation allows the control to be penalized only in the null space direction and only allow torque error when approaching a singularity. In contrast the Moore-Penrose pseudoinverse steering law regularizes the term $B_w(x)B(x)^T$ which adds torque error at every timestep and it is not guaranteed to clear the singularity in some cases.

Results

In this experiment, we set a desired attitude to the identity quaternion. The initial quaternion is sampled from a Gaussian with a 1σ standard deviation of 11 degrees while the initial angular velocity is also sampled from a Gaussian with a standard deviation of 0.2 rad/s. The rotation axes are randomly sampled and normalized. The simulation was performed at 10 Hz for 100 seconds. We present the result of two randomized initial condition that cause a singularity in the CMG array throughout the simulation. We compare our control barrier function approach to a Moore-Penrose pseudoinverse method with different size regularization parameters. The results are shown in Fig. 75, 76, and 77. Our CBF approach avoids the singularity encountered in pseudoinverse

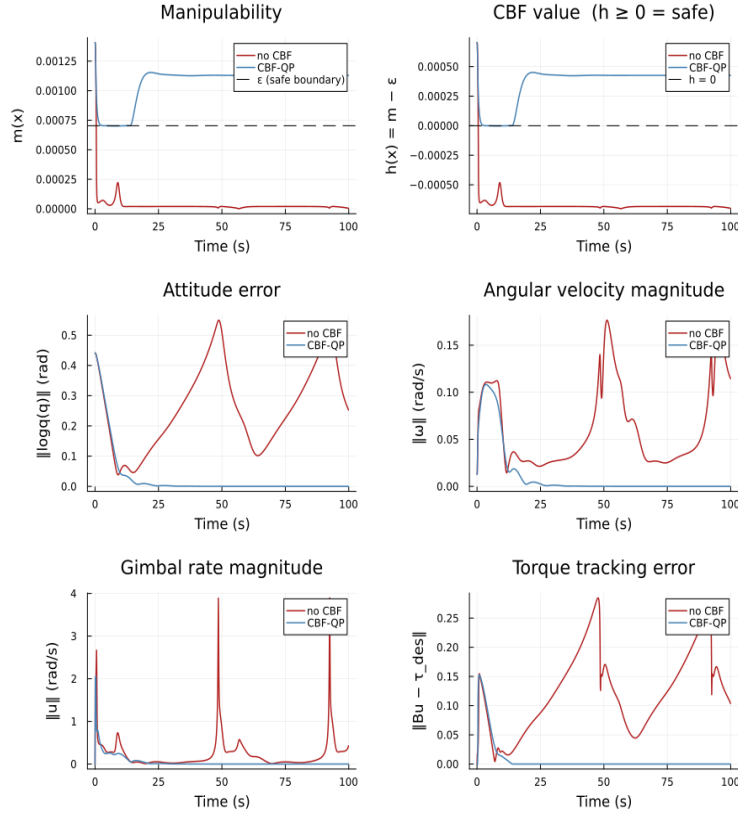


Figure 75: Test 1: CBF vs Regularized Pseudo Inverse Steering Law Comparison (with a regularizer $1 \times 10^{-3}I$)

approach within the first 10 seconds of the simulation. This singularity led the pseudoinverse approach to command high gimbal rate commands that caused the system to never converge to the desired attitude. Adding more torque error by increasing the regularizer can mitigate high gimbal rates, but it still may not reach the desired attitude as seen in Figure 77. However, there are certain initial conditions in which the pseudoinverse approach converged to the desired attitude at the expense of oscillating the gimbal rate commands through the singularity which is not ideal to run on hardware. This example is shown in Fig. 76

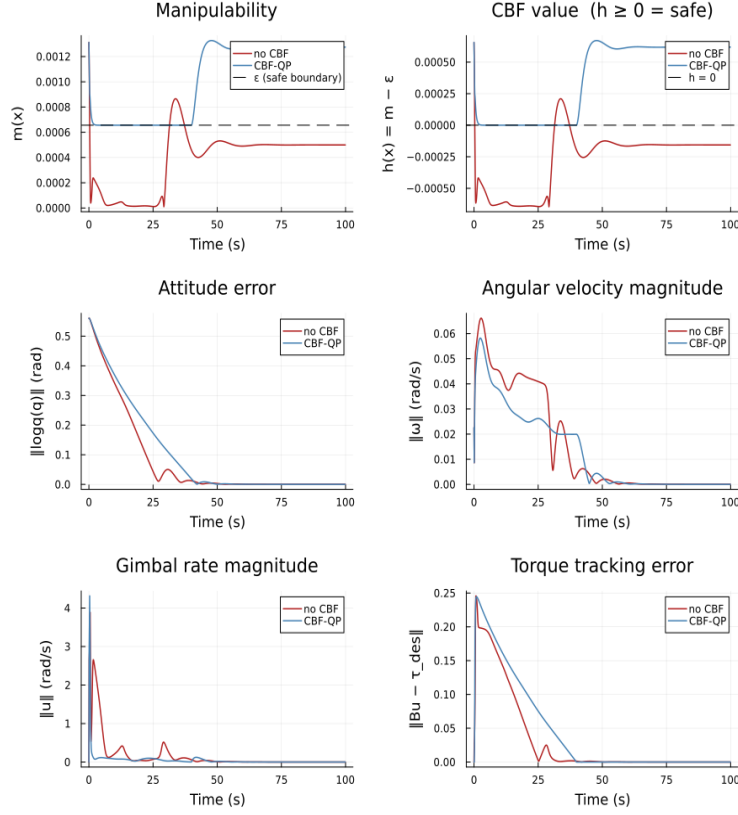


Figure 76: Test 2: CBF vs Regularized Pseudo Inverse Steering Law Comparison (with a regularizer $1 \times 10^{-3}I$)

Conclusions

Our optimization-based steering law uses a control barrier function to ensure the CMG actuator jacobian stays well-conditioned throughout the trajectory by avoiding the singularity CMG configurations. This conservative approach comes with a tradeoff as providing a formal guarantee of singularity avoidance may decrease attitude tracking performance in some cases. However, it alleviates the need for a parameter-dependent psuedoinverse that is not guaranteed to work at all initial conditions. Looking ahead, we aim to explore different control barrier functions that may be more robust numerically compared to the manipulability index as well as tuning the α in the CBF constraint. We also plan on incorporating a look-ahead trajectory to the QP and analyze the performance. Another interesting direction is posing the trajectory optimization problem in the gimbal rate space and solve the nonconvex obstacle avoidance problem (singularities as the obstacles in the gimbal rate space).

Appendix

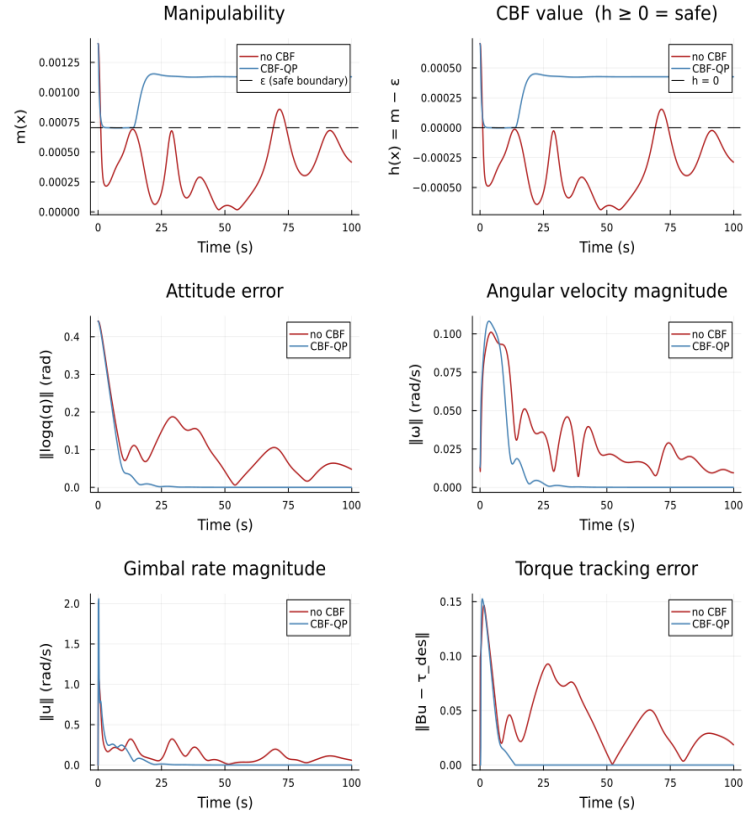


Figure 77: Test 1: CBF vs Regularized Pseudo Inverse Steering Law Comparison (with a regularizer $1 \times 10^{-2} I$)

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